

ME1103

Principles of Mechanics and Materials

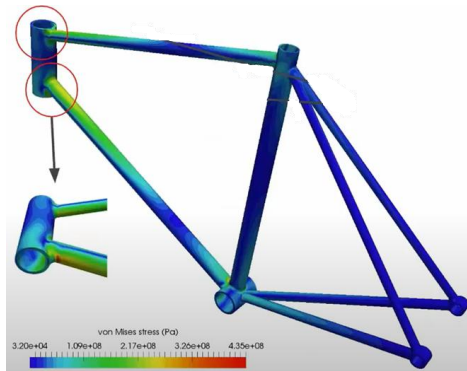
INTERNAL FORCE AND MOMENT IN SLENDER STRUCTURES

Learning Outcomes

- Understand internal force and moment in structural members and using method of sections to determine them at specific location.
- Understand the characteristics of beam, and the internal shear and moment induced by the external lateral loads.
- Understand the relationships between the external loads, internal shear and moment, and hence able to plot the shear and moment diagrams.

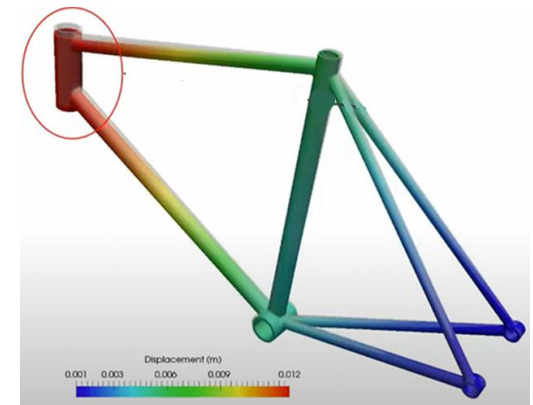
Failure in Mechanical Structures

- So far, we have been solving the internal forces at the connecting joints of different members in an assembly. They are essentially the external forces acting on the individual members that induced stresses within the body.
- It is also important to determine the internal forces and its intensity (*stress*) in the members. They are the forces that hold the material together in the body and causes infinitesimal deformation (*strain*) in the body.
- Internal forces are related two common failures in structures, namely:



High stress level compared to the material strength.

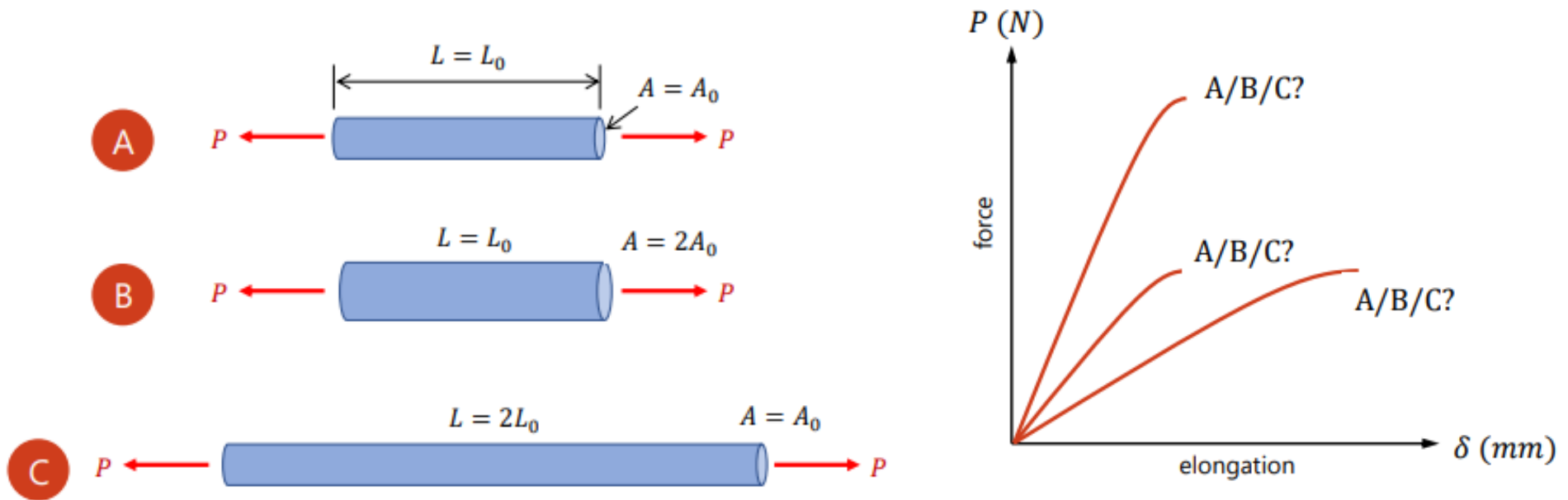
Structural analysis of bike frame



Excessive deformation of the bike frame.

Concept of Stresses and Strains – Normal stress

- Suppose we pull three different specimens of different dimensions but made of the same material.
- We will get three different curves for the force vs elongation plot.



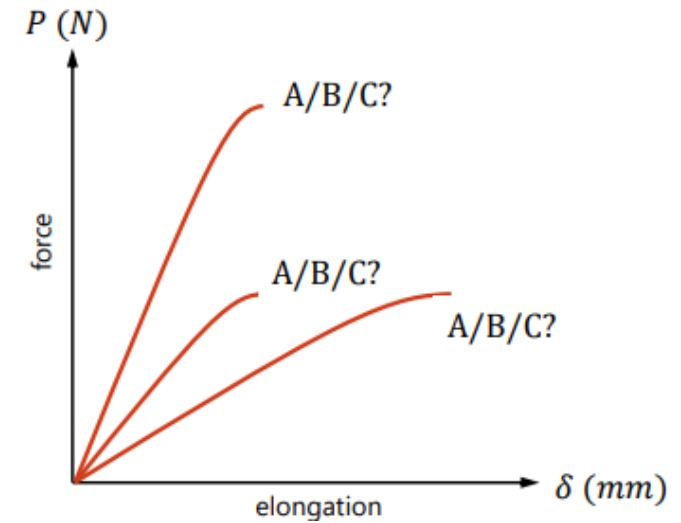
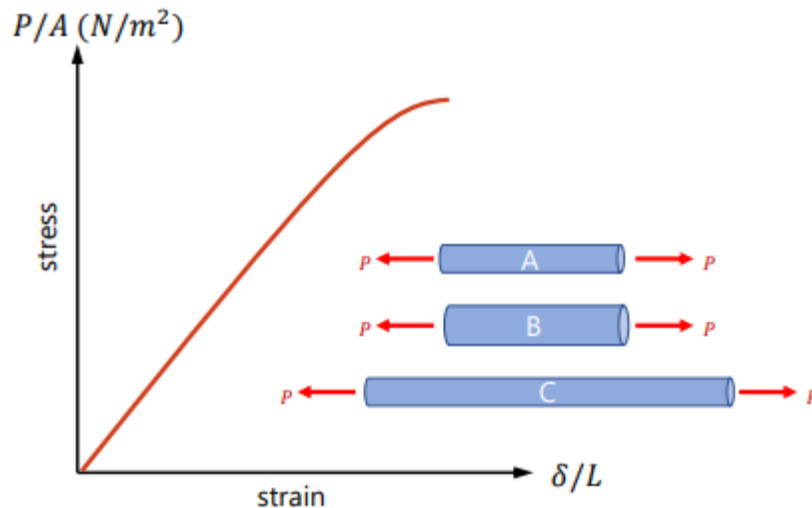
Tensile test results for A, B and C

Concept of Stresses and Strains – Normal stress...

➤ Key observations:

- Larger cross-section area A can carry larger load P .
- Longer length L can sustain larger elongation δ .

➤ So what if we plot (P/A) vs (δ/L) ?



Stress is the intensity of force acting over a unit area.

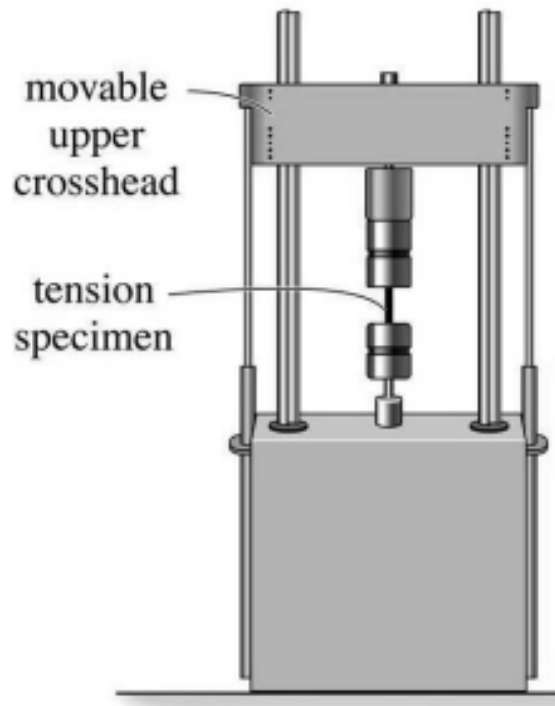
$$\sigma = \frac{P}{A}, \quad \left[\frac{N}{m^2} \text{ or Pascal (Pa)} \right]$$

Strain is the ratio of deformation w.r.t. the specimen dimensions.

$$\varepsilon = \frac{\delta}{L}, \quad [\text{dimensionless}]$$

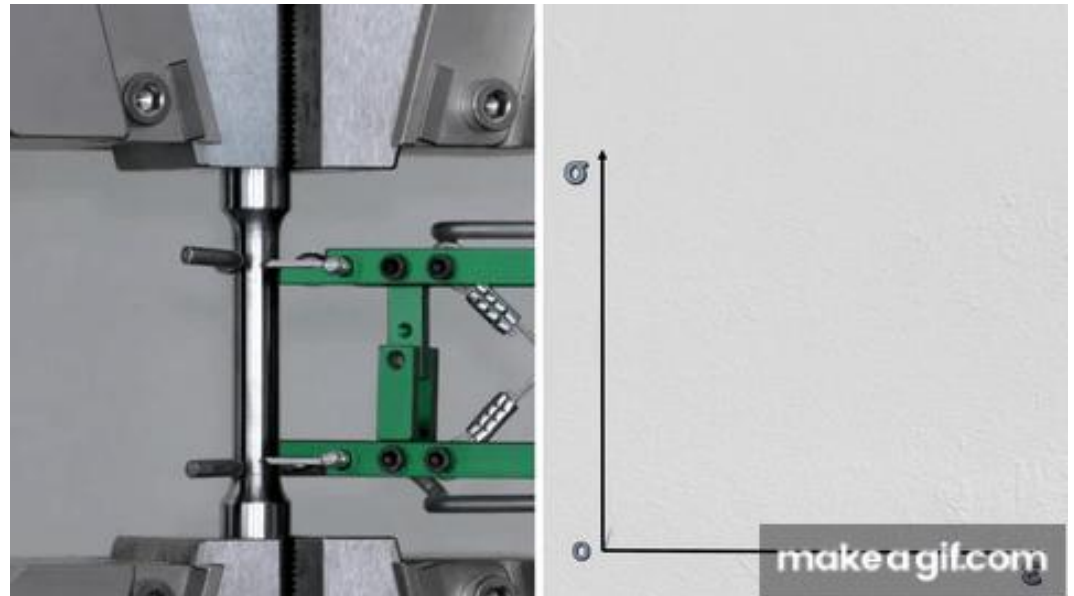
Concept of Stresses and Strains – Normal stress...

- Uniaxial tensile test is commonly performed to determine the mechanical properties of materials.



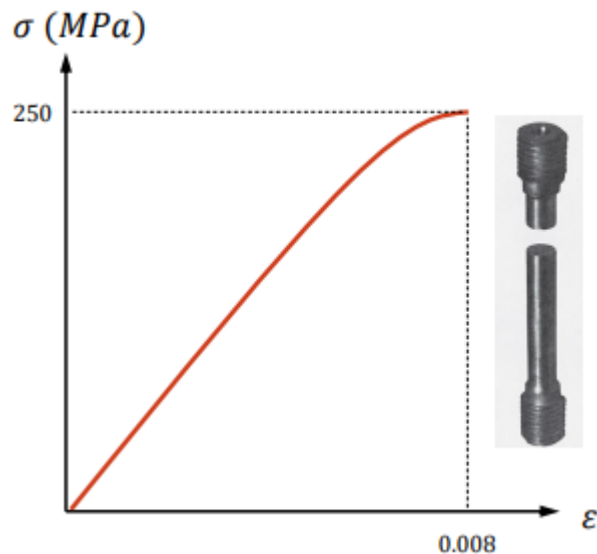
Universal Testing Machine

Tensile test of a ductile material

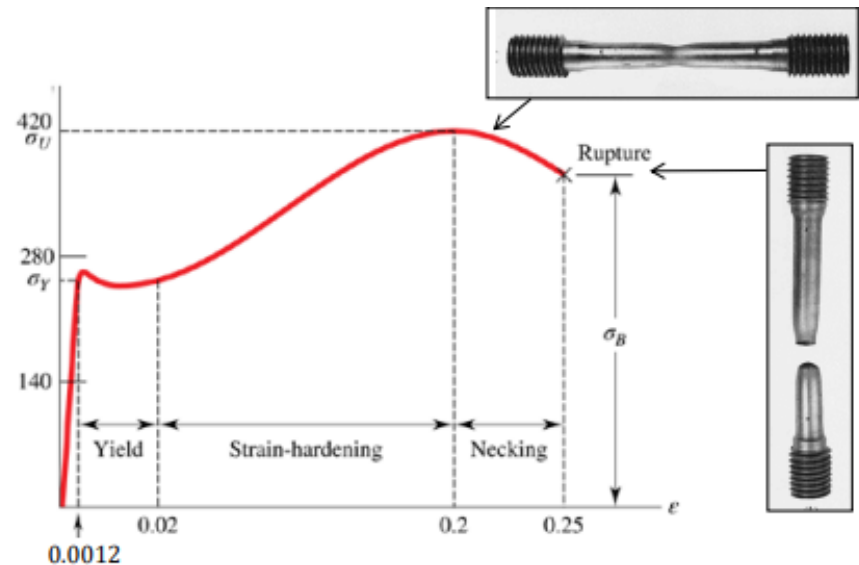


Concept of Stresses and Strains – Normal stress...

- In summary, every material has a unique stress vs strain relation that reflects its mechanical properties.
- Broadly, materials can be classified as brittle or ductile.



Brittle material, e.g. cast iron



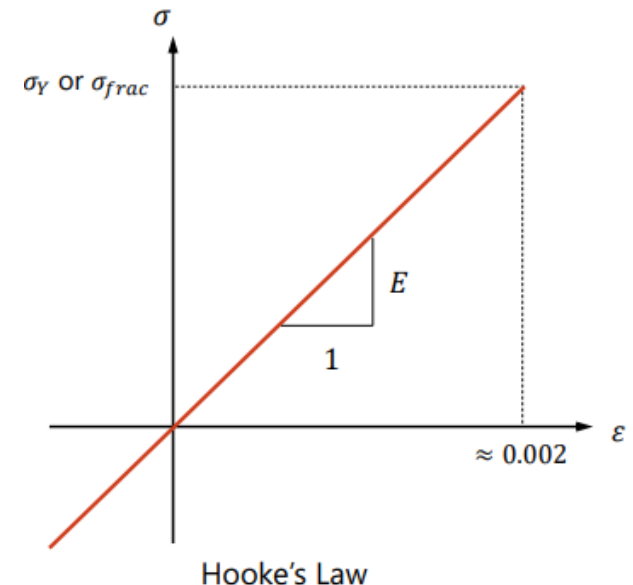
Ductile material, e.g. aluminium

Concept of Stresses and Strains – Young's modulus

- For most materials, there exist a regime in which the materials behave elastically. That is, the structure will restore to its original undeformed dimensions if the load is removed.
- For brittle material, the elastic limit is usually defined when the strain is less than 0.002.
- For ductile material, it is defined when the stress level is less than the yield strength of the material.
- Within the elastic regime, the stress and strain are related linearly by the Hooke's Law,

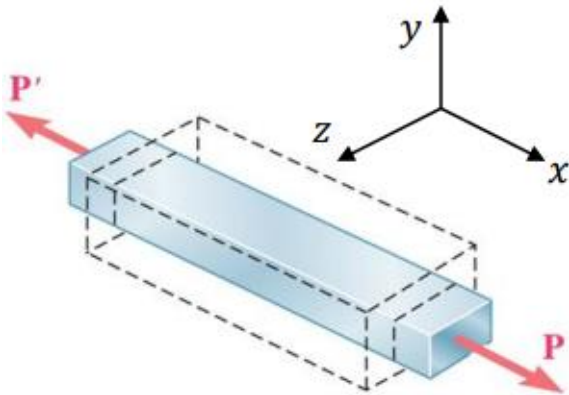
$$\sigma = E\varepsilon$$

where the gradient E is known as the Young's modulus.



Concept of Stresses and Strains – Poisson's ratio

- When a rod is stretched (or compressed) axially, one can also observe a corresponding contraction (or expansion) in the cross-section dimensions.



- This implies that normal (axial) stress not only gives rise to normal strain in the loading direction, i.e.

$$\epsilon_x = \frac{\sigma}{E}$$

- It also gives rise to transverse strains in the perpendicular directions, i.e.

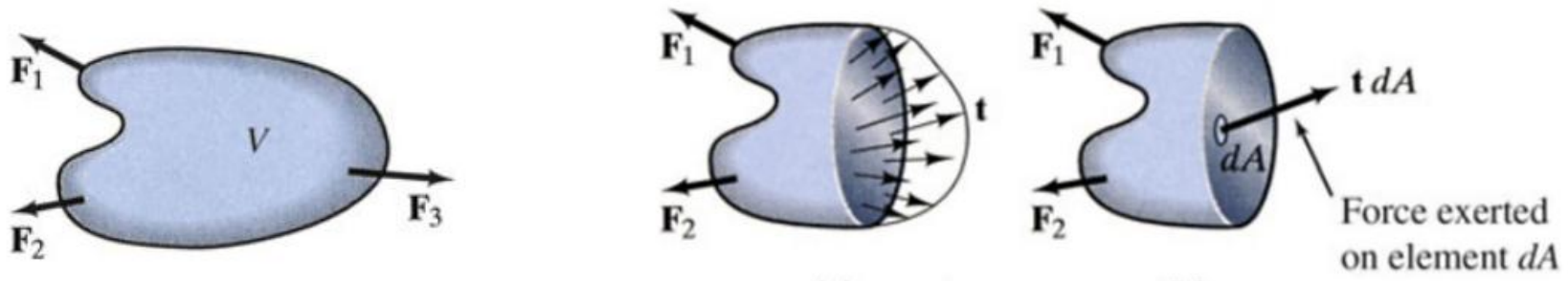
$$\epsilon_y = -\nu\epsilon_x; \quad \epsilon_z = -\nu\epsilon_x$$

where ν is known as the Poisson's ratio of the material.

Note: This Poisson's effect is NOT due to the conservation of volume, unless the material is incompressible, e.g. rubber, where $\nu = 0.5$.

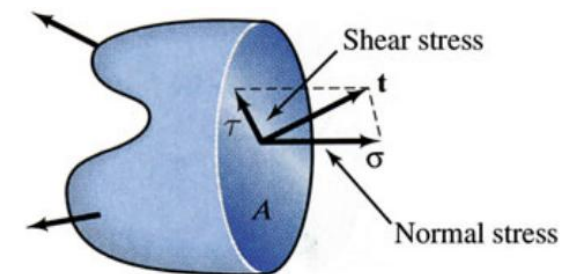
Concept of Stresses and Strains – 3D stresses

- Consider a body that is loaded by multiple forces in arbitrary directions.
- Slicing it at a given plane will reveal the internal forces at the cut surface. In general, the internal force acting on an infinitesimal area will not be normal to cut surface, and they are not uniform over the surface..



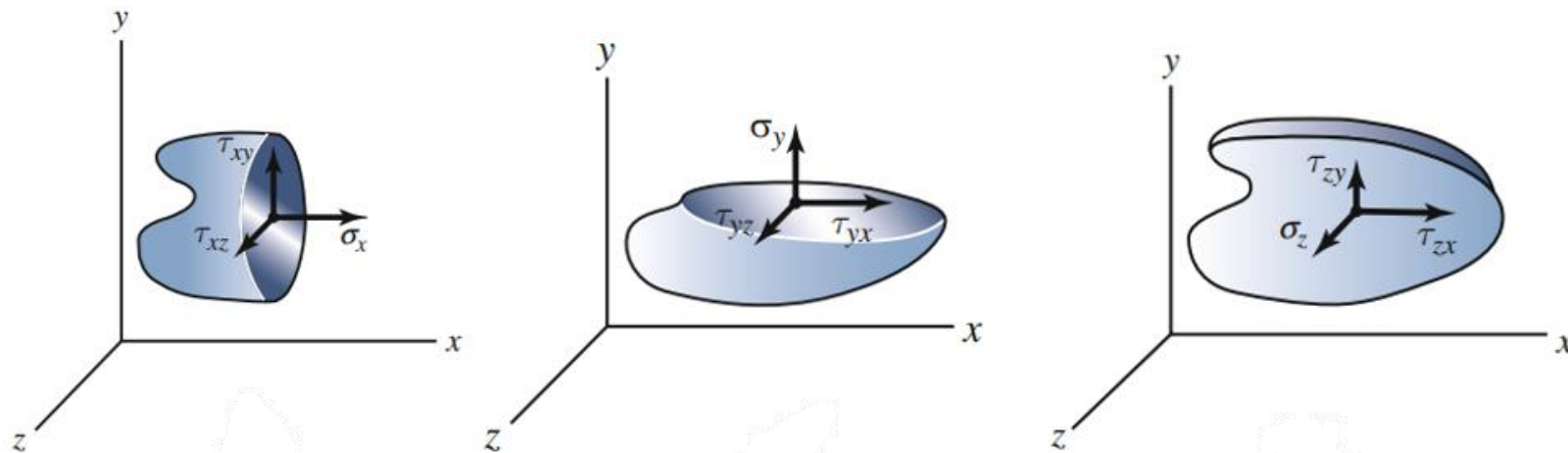
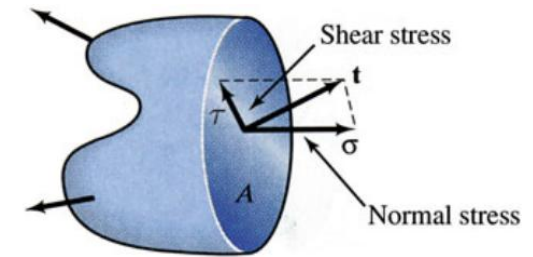
- This implies that there could be two types of stresses that can be defined on a given surface, namely:

- Normal stress σ – perpendicular to the cut surface.
- Shear stresses τ – tangential to the cut surface.



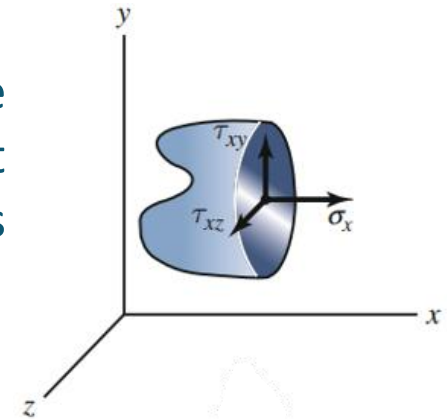
Concept of Stresses and Strains – 3D stresses...

- Notice that normal stress is uniquely defined by the cut surface. But this is not the case for shear stress, since it can be defined in any direction as long as it is tangential to the surface.
- However, we can introduce a coordinate system (cartesian coordinate) to formalize its definition consistently. As such, we can define three orthogonal cutting planes that are normal to the x-, y- and z-axes.

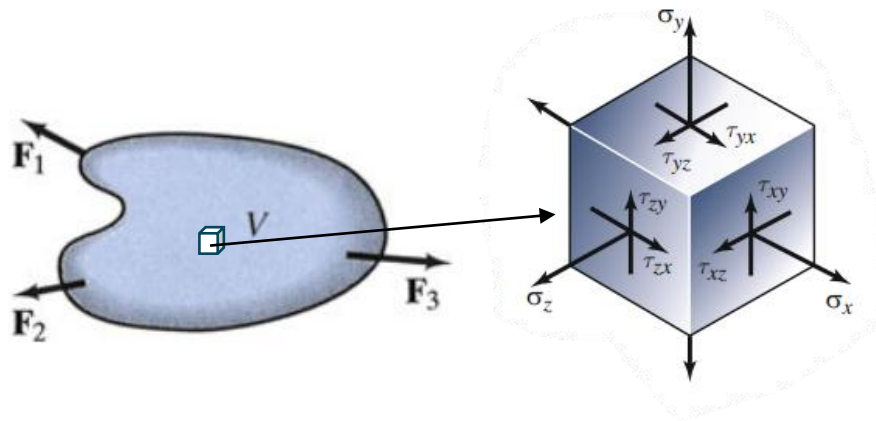


Concept of Stresses and Strains – 3D stresses...

- With the coordinate system, the shear stresses can now be uniquely identified with two indices in the subscript. The first index denotes the cutting plane while the second index shows the direction of the applied shear force.



- To completely describe the stress state at a given point, we isolate an infinitesimal cubical element from the body.



- Note however that there are only 6 stress components, namely:

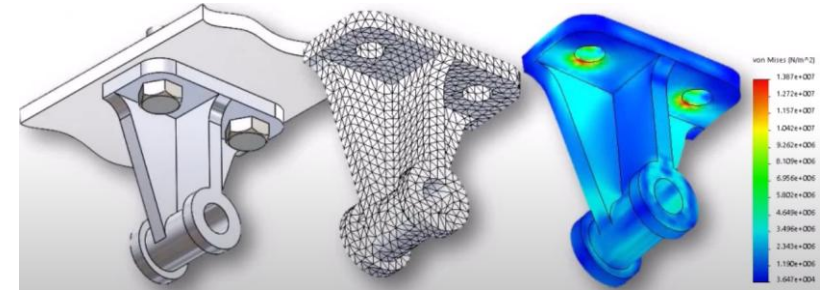
$$\sigma_x, \sigma_y, \sigma_z, \tau_{xy}, \tau_{xz} \text{ and } \tau_{yz}$$

because the shear stresses are related

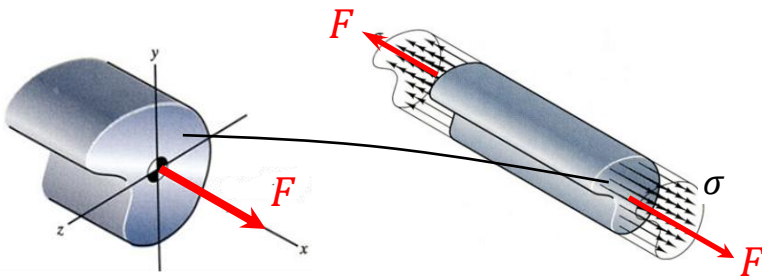
$$\tau_{xy} = \tau_{yx}; \tau_{xz} = \tau_{zx}; \tau_{yz} = \tau_{zy}$$

Slender Structures

- In general, stress induced within the structures can be quite complex and cannot be evaluated analytically, primary due to the complexity of the structure.
- However, for a slender rod that has small cross-sectional dimensions compared to its length, i.e. $L/d > 10$, analytical formulas can be derived.



Axially loaded rod/bar



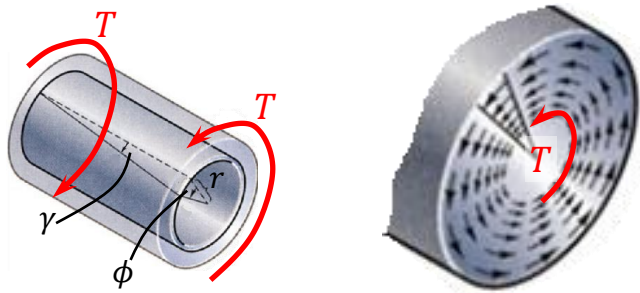
The induced uni-axial stress and strain are given by:

$$\sigma_x = \frac{F}{A}; \quad \epsilon_x = \frac{d\delta}{dx} = \frac{F}{EA}$$

where A is the cross-section area of the rod, E is the Young's modulus of the material, and δ is the deformation of the rod due to the applied force F .

Slender Structures...

Twisting of circular shaft



The induced shear stress and strain are given by:

$$\tau = \frac{Tr}{J}; \quad \gamma = \frac{d\phi}{dx} = \frac{T}{GJ}$$

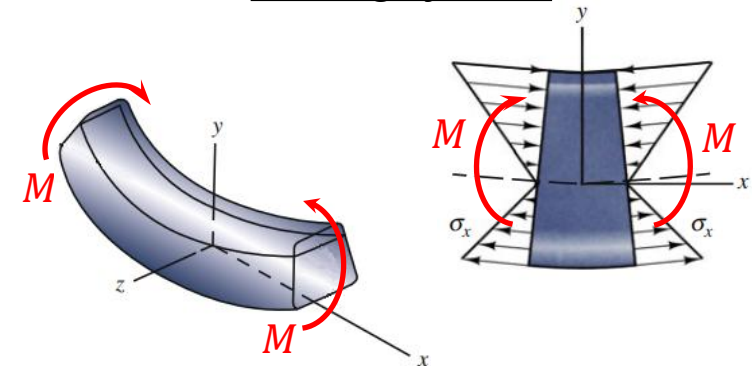
where J is the Polar 2nd moment of area of the rod, G is the shear modulus of the material, and ϕ is the angle of twist of the rod due to the applied torque T .

The induced normal stress and lateral deflection of beam v are given by:

$$\sigma_x = -\frac{My}{I}; \quad \frac{d^2v}{dx^2} = \frac{M}{EI}$$

where I is the 2nd moment of area w.r.t. the direction of the bending moment, and v is the lateral deflection of the rod due to the applied moment M .

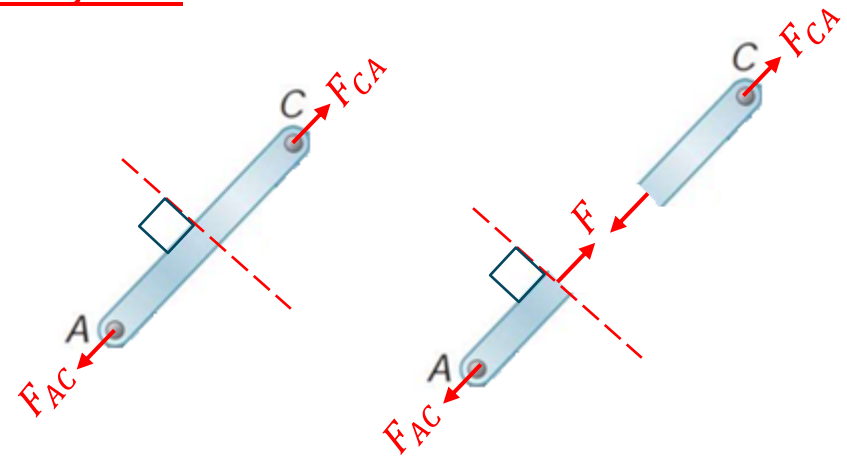
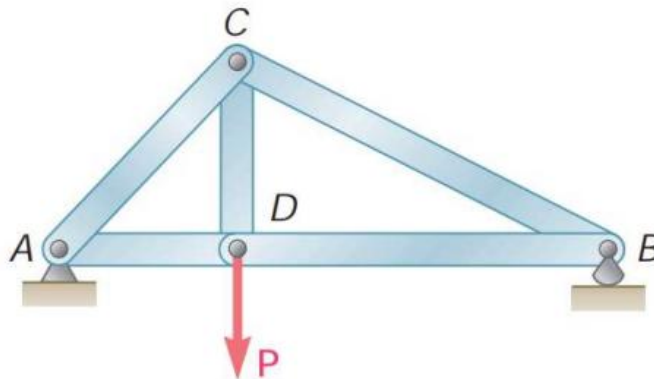
Bending of beam



Stresses and strains (deformation) induced in slender structures are related to the internal force and moment in the body, which in turns come from the external loads applied onto it.

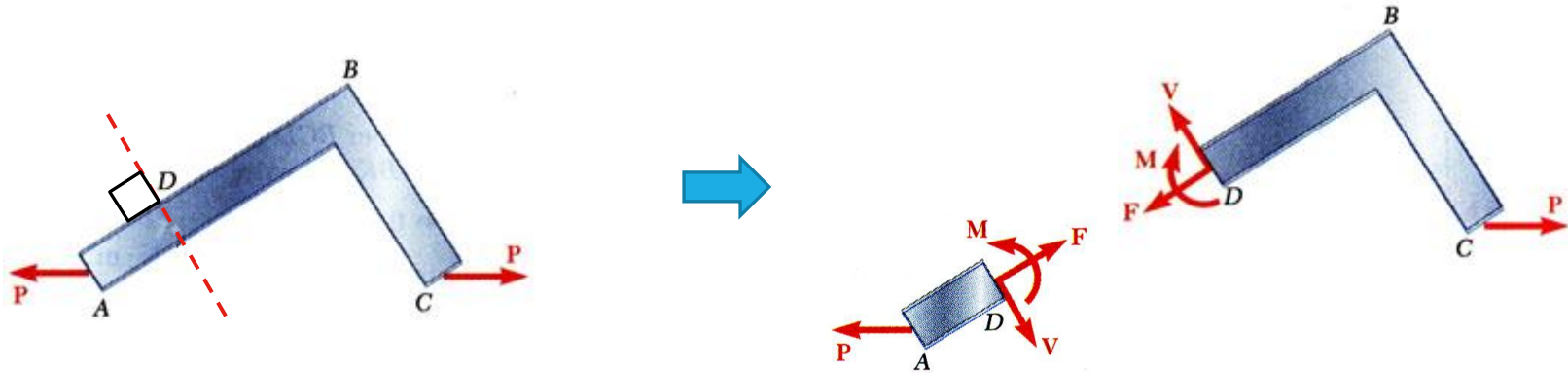
Internal Forces in Truss

- In general, internal force and moments can vary along the slender body, depending on the loading and boundary (supporting) conditions.
- They can only be determined if the body is cut at the specific position and isolated as *FBD* in its statics analysis. This is essentially the method of section.
- Recalled that members in truss are axially loaded rod where the member force is uniform along its axis with the same force.
- When the member is cut at a plane that is perpendicular to its axis, the internal force is often called the axial force or normal force.

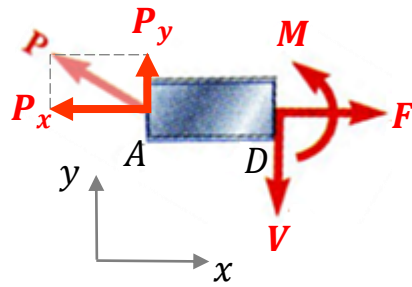


Internal Forces and Moment

- On the other hand, if the member is not straight, or is a multiple-force member, then two internal forces and one moment are required at the cut surface to maintain the static equilibrium.



- Static equilibrium means

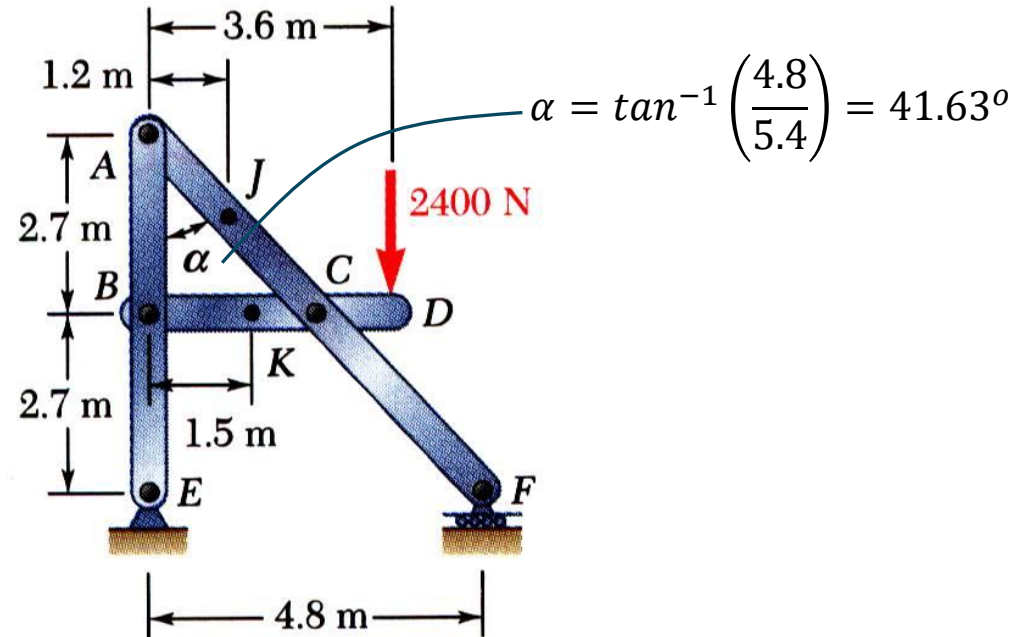


$$F = P_x \text{ (Normal force)}$$

$$V = P_y \text{ (Shear force)}$$

$$M = -\vec{r}_{A/D} \times \vec{P} \text{ (Bending moment)}$$

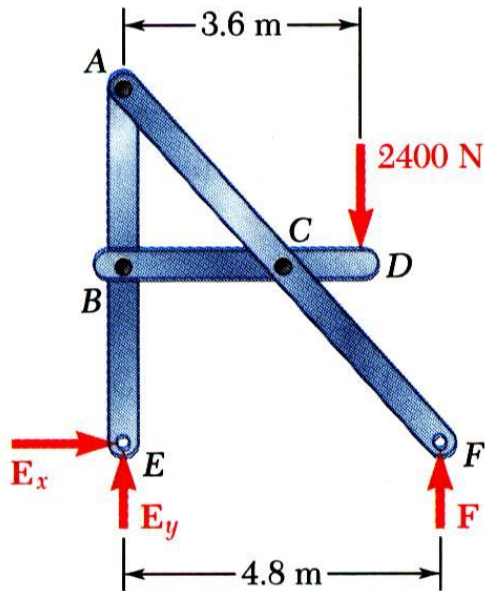
Example 5.1



Determine the internal forces and moments:

- (a) in member ACF at point J , and
- (b) in member BCD at K .

Example 5.1...



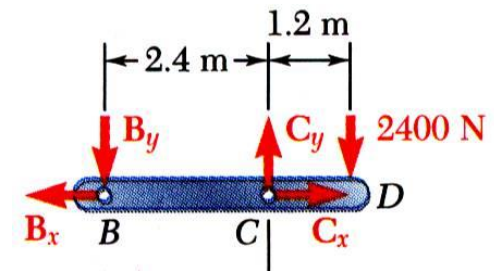
Based on the *FBD* of the entire frame, we get

$$\sum M_E = 0, \quad F(4.8) - (2400)(3.6) = 0 \Rightarrow F = 1800 \text{ N}$$

Consider member *BCD*, we have

$$\sum M_B = 0, \quad -(2400)(3.6) + C_y(2.4) = 0$$

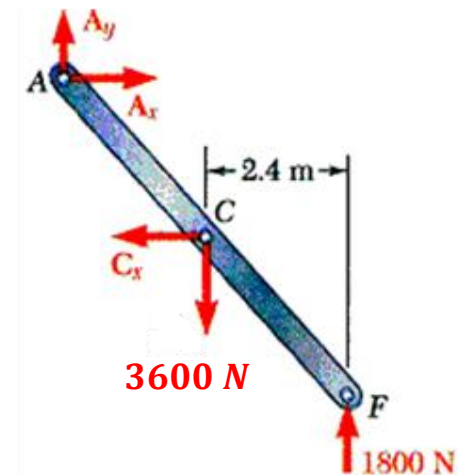
$$\Rightarrow C_y = 3600 \text{ N}$$



To determine C_x , we consider member *ACF*, i.e.

$$\sum M_A = 0, \quad -(3600)(2.4) + (1800)(4.8) - C_x(2.7) = 0$$

$$\Rightarrow C_x = 0$$



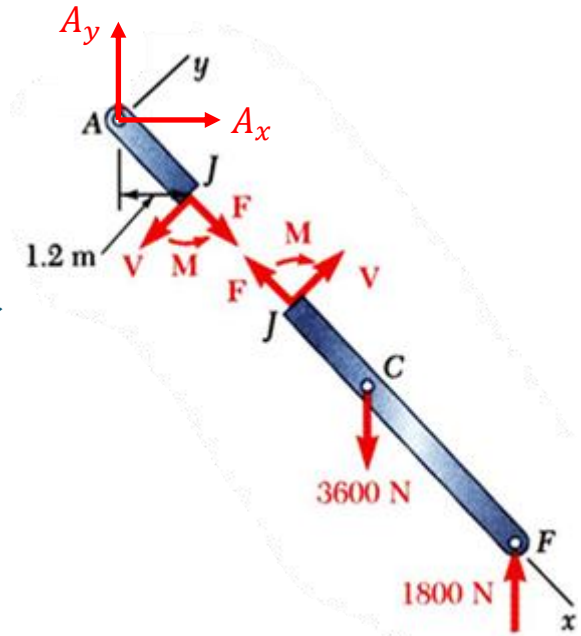
Example 5.1...

(a) Consider member *ACE* and cut it at *J*. The internal forces at *J* are represented by equivalent force-couple pair. And static equilibrium of section *JCF*, gives

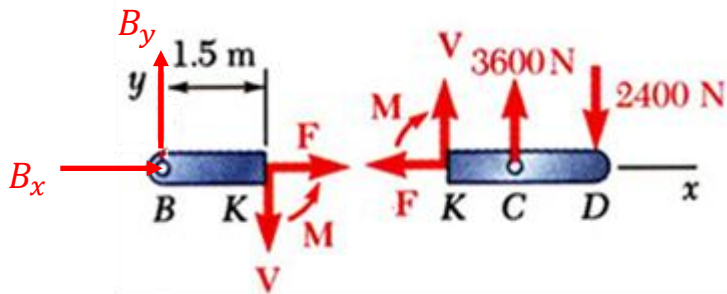
$$\sum M_J = 0, M + 3.6(1.2) - (1.8)(3.6) = 0 \Rightarrow M = \mathbf{2.16 \text{ kN.m}}$$

$$\sum F_x = 0, F + (1.8 - 3.6)\cos 41.7^\circ = 0 \Rightarrow F = \mathbf{1.334 \text{ kN}}$$

$$\sum F_y = 0, V + (1.8 - 3.6)\sin 41.7^\circ = 0 \Rightarrow V = \mathbf{1.197 \text{ kN}}$$



(b) Similarly cut member *BCD* at *K*.



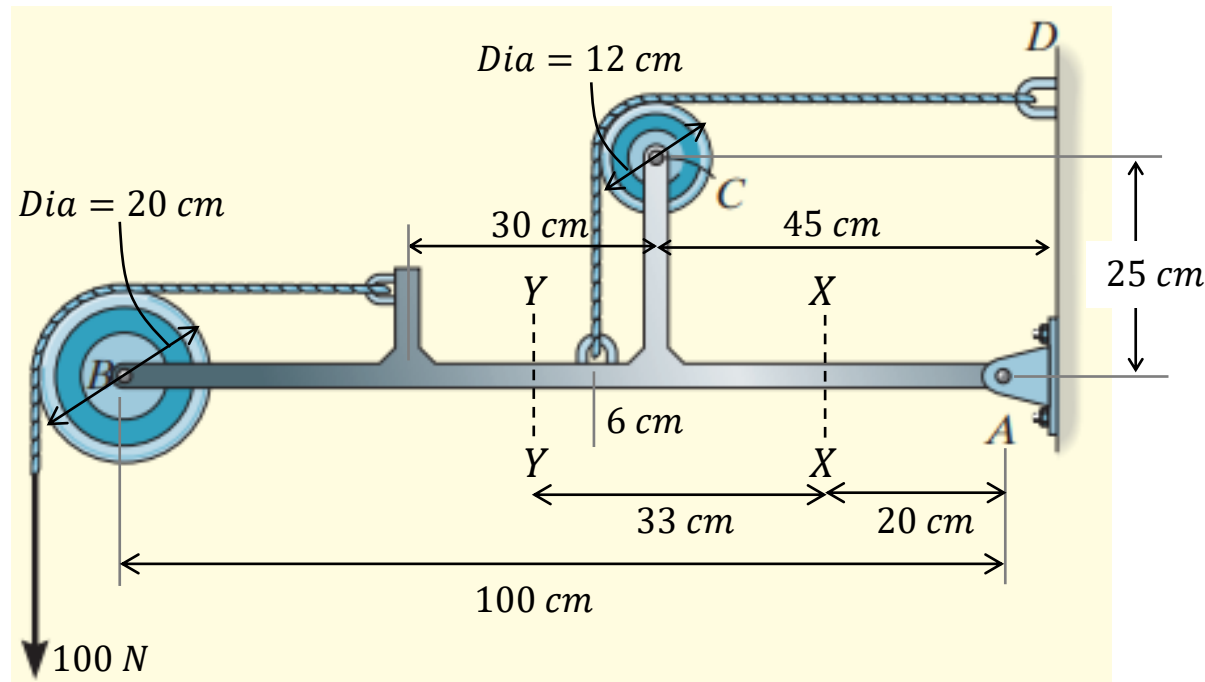
$$\sum M_K = 0, M + 2.4(2.1) - 3.6(0.9) = 0$$

$$\Rightarrow M = \mathbf{-1.8 \text{ kN.m}}$$

$$\sum F_y = 0, V + 3.6 - 2.4 = 0 \Rightarrow V = \mathbf{-1.2 \text{ kN}}$$

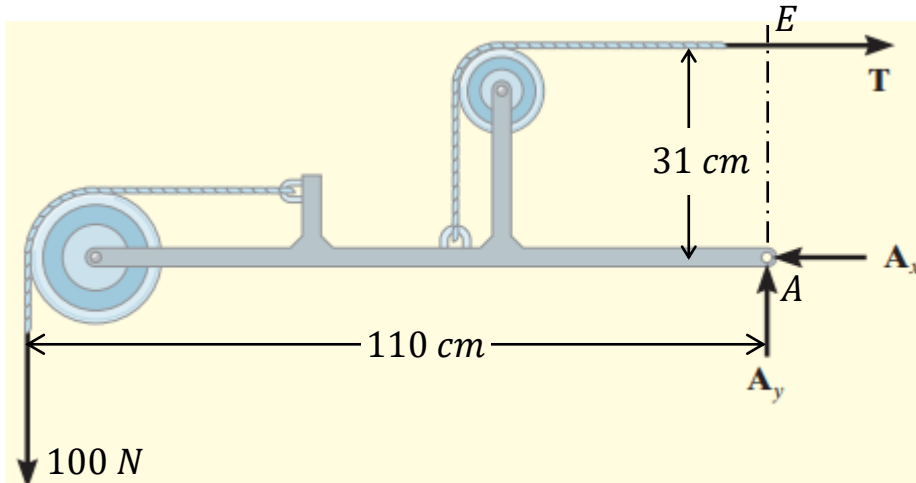
Example 5.2

For the given frame shown in the figure, determine the internal forces at the cut sections defined by X-X, and Y-Y.



Example 5.2

Consider the *FBD* of the entire frame.



$$\sum F_y = 0, \quad A_y = 100 \text{ N}$$

$$\sum M_E = 0, \quad 100(110) - A_x(31) = 0$$

$$\Rightarrow A_x = 354.8 \text{ N}$$

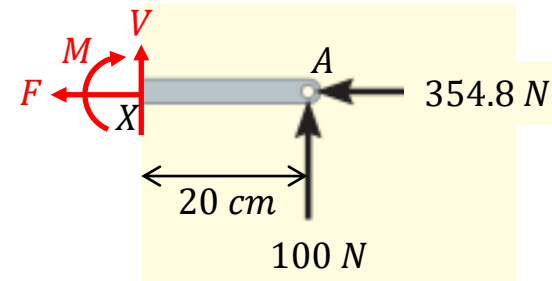
$$\sum F_x = 0, \quad T = 354.8 \text{ N}$$

$$\sum F_x = 0, \quad F = -354.8 \text{ N}$$

$$\sum F_y = 0, \quad V = -100 \text{ N}$$

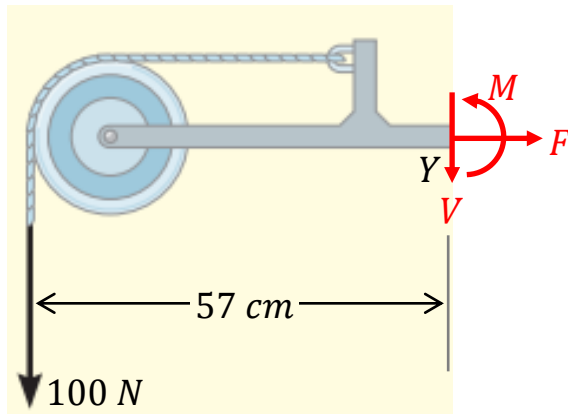
$$\sum M_X = 0, \quad M = 100(0.2) = 20 \text{ N.m}$$

With the reaction at A, consider the *FBD* of the segment XA.



Example 5.2

Next, consider the *FBD* of the left segment cut by Y-Y.



$$\sum F_x = 0, \quad F = 0 \text{ N}$$

$$\sum F_y = 0, \quad V = -100 \text{ N}$$

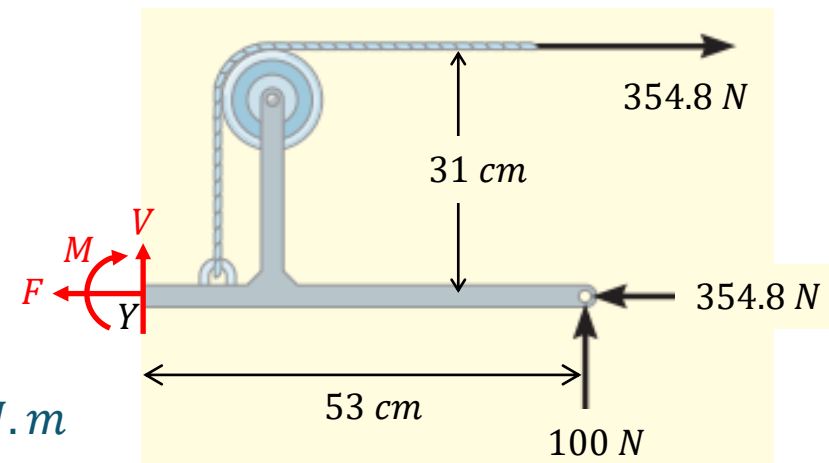
$$\sum M_Y = 0, \quad M = -100(0.57) = -57 \text{ N.m}$$

Or using the *FBD* of the right segment cut by Y-Y.

$$\sum F_x = 0, \quad F = 0 \text{ N}$$

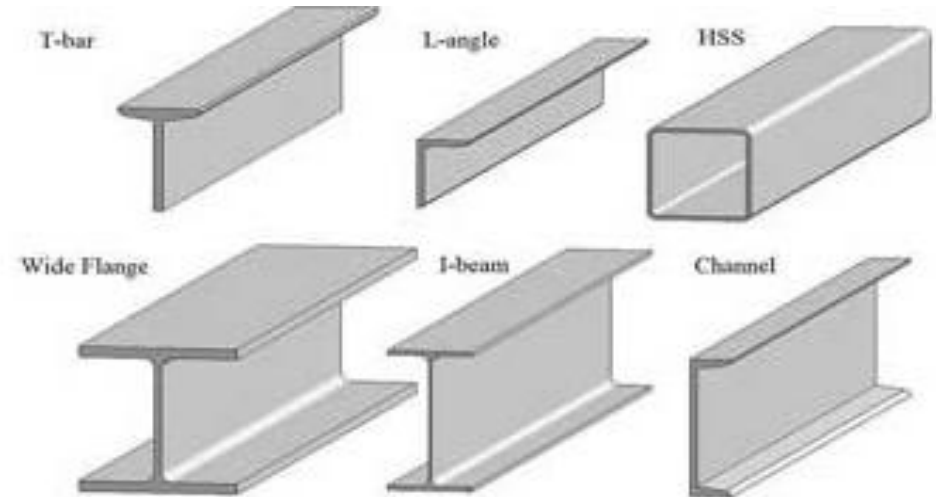
$$\sum F_y = 0, \quad V = -100 \text{ N}$$

$$\sum M_Y = 0, \quad M = 100(0.53) - 354.8(0.31) = -57 \text{ N.m}$$



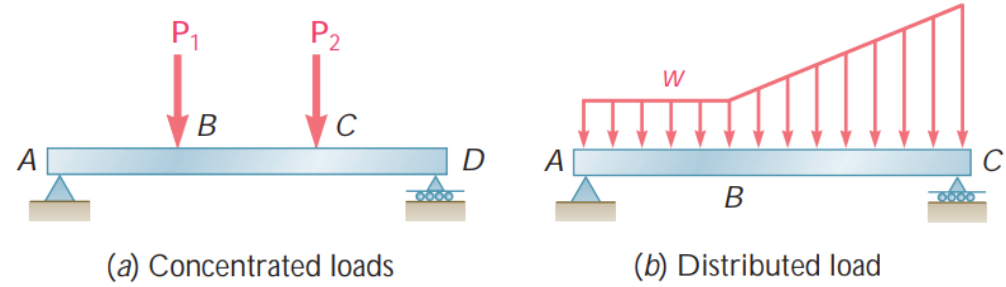
Beam

- Beams are usually long, slender and straight bars that are designed to support lateral loads that are perpendicular to the axis of the beam.
- They are the most common type of structure components, particularly in buildings and bridge construction applications.
- They are used because of their superior strength to weight ratio.

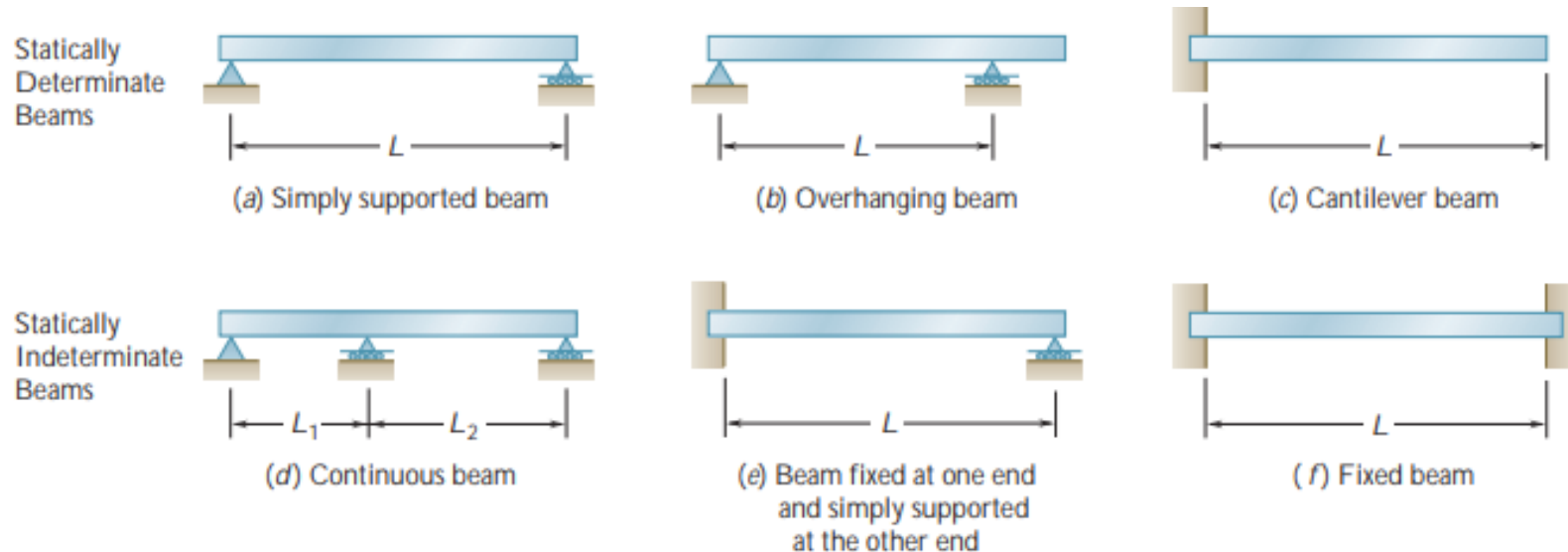


Beam...

- Beams can be subjected to concentrated (point) loads or distributed load (load per unit length).



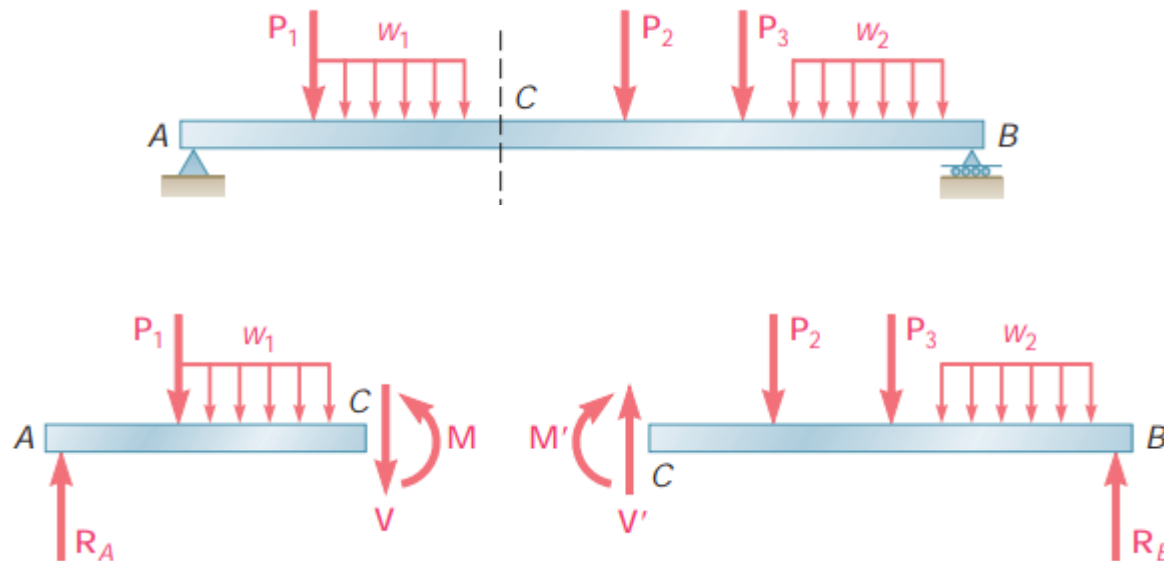
- Beams are also classified according to how they are supported.



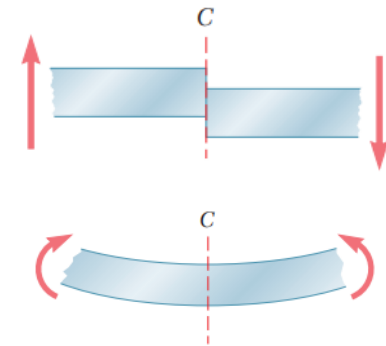
Analysis of Beam

➤ Typically, analyzing a beam requires:

- First, determining the shearing forces and bending moments in the structure, and then
- Evaluate the deformation and/or stresses induced in the beam for a given cross-section profile.

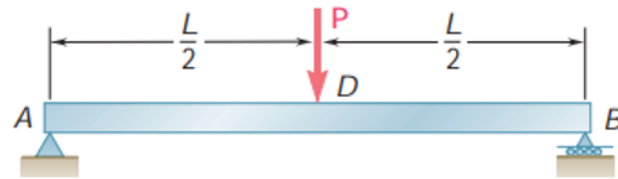


Sign convention

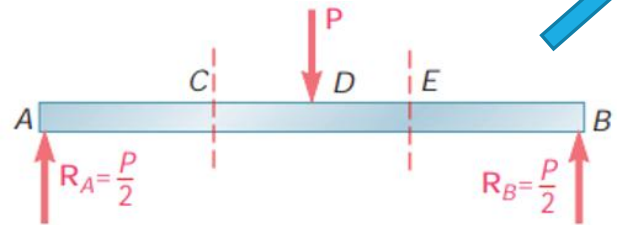


Shear and Bending Moment Diagrams

- Consider the simply-supported beam with a point load as shown.

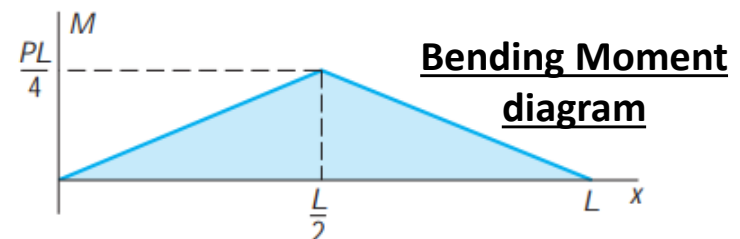
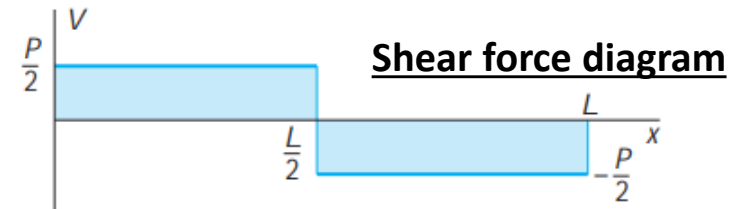
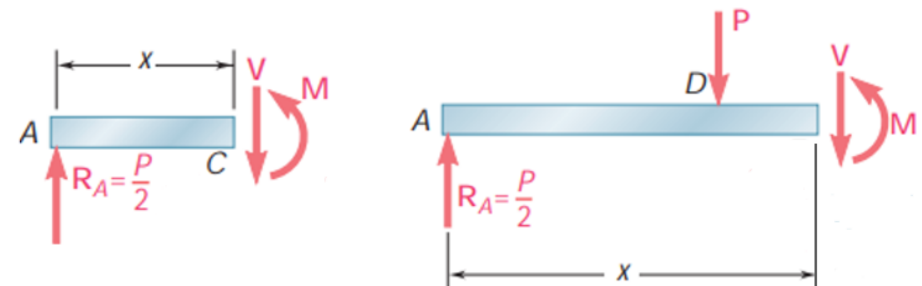


Simply-supported beam



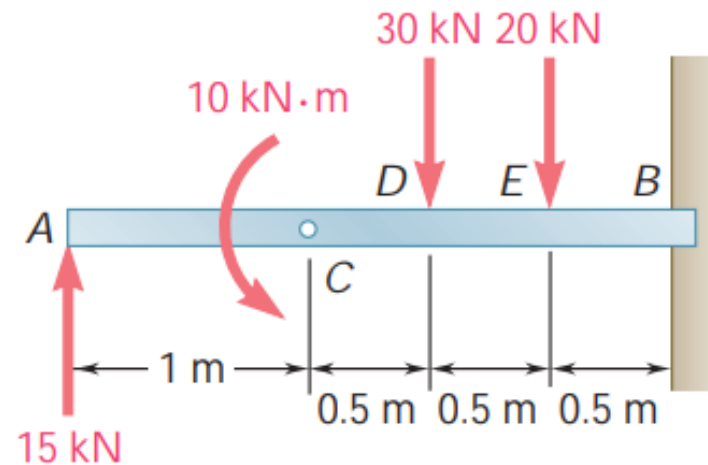
- Determine reaction forces by static analysis.
- Identify distinct segments with different loading conditions.

- Applied method of section to each segment to determine the shear and bending moment profile.



Example 5.3

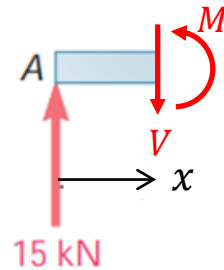
For the beam and loading shown, sketch the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.



Example 5.3...

At a point slightly from A , the shear force must balanced the applied load, i.e. $V = 15 \text{ kN}$.

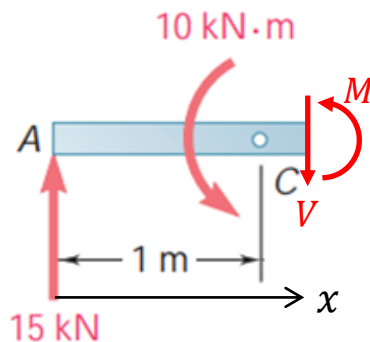
Consider a cut at segment AC .



$$\sum F_y = 0, \quad V = 15$$

$$\sum M_x = 0, \quad M = 15x$$

Consider a cut at segment CD .



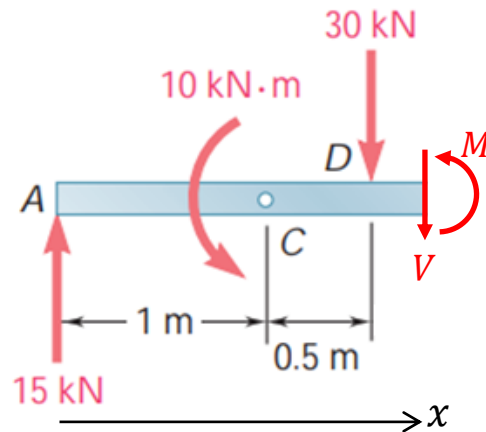
$$\sum F_y = 0, \quad V = 15$$

$$\sum M_x = 0, \quad M + 10 - 15x = 0$$

$$\Rightarrow M = 15x - 10$$

Example 5.3...

Consider a cut at segment *DE*.

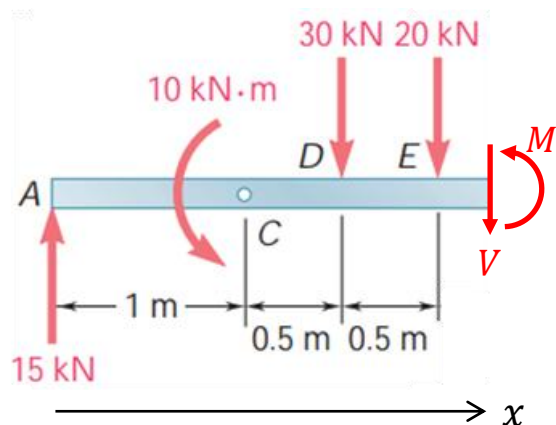


$$\sum F_y = 0, \quad V + 30 - 15 = 0 \Rightarrow V = -15 \text{ kN}$$

$$\sum M_x = 0, \quad M + 30(x - 1.5) + 10 - 15x = 0$$

$$\Rightarrow M = -15x + 35$$

Consider a cut at segment *EB*.



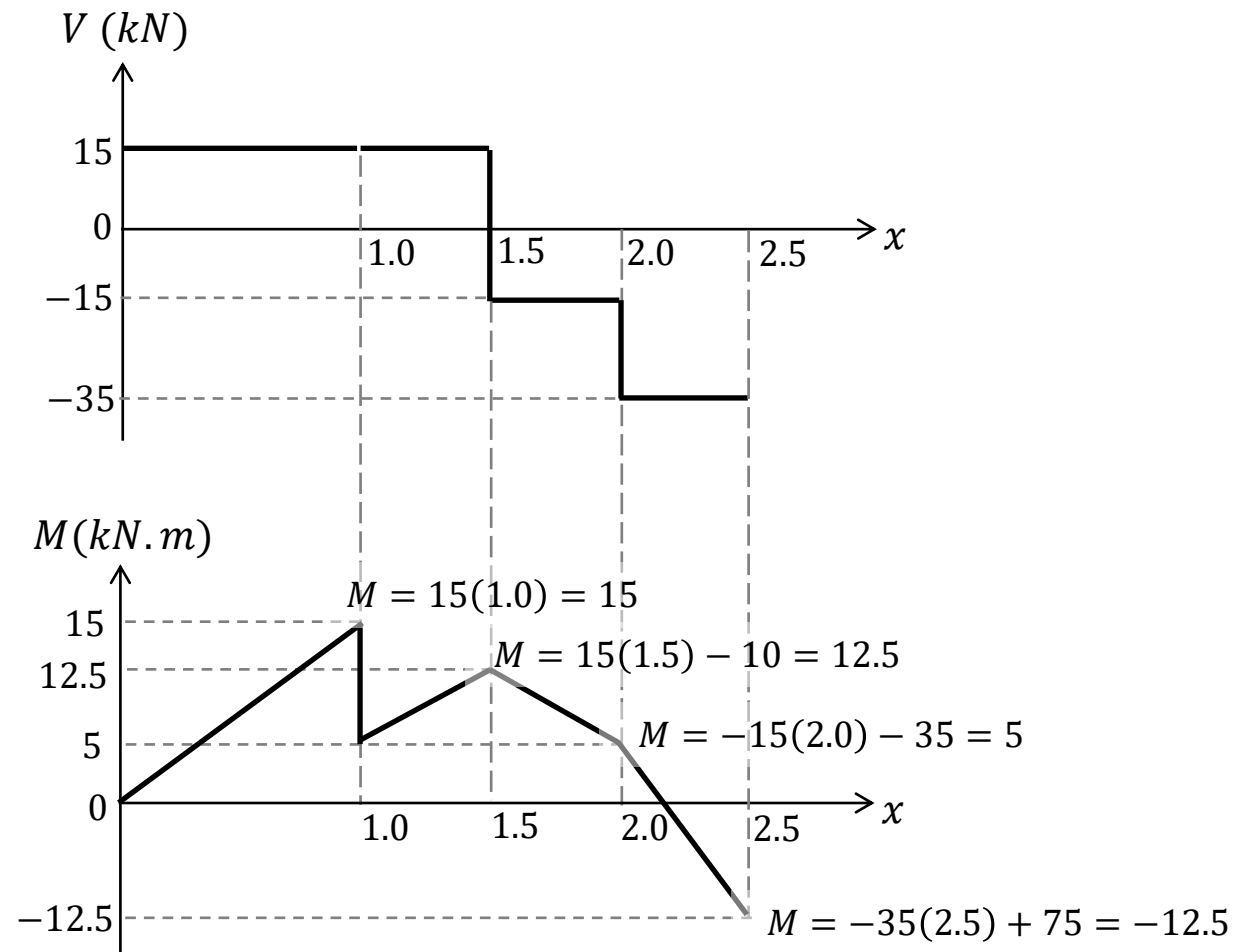
$$\sum F_y = 0, \quad V + 30 + 20 - 15 = 0 \Rightarrow V = -35 \text{ kN}$$

$$\sum M_x = 0, \quad M + 30(x - 1.5) + 20(x - 2) + 10 - 15x = 0$$

$$\Rightarrow M = -35x + 75$$

Example 5.2...

Finally, plotting all the expressions for the shear force and bending moment along the beam, gives



Reaction Forces with Distributed Loads

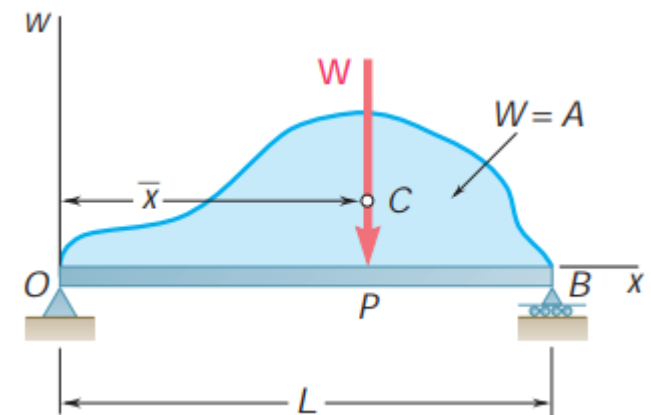
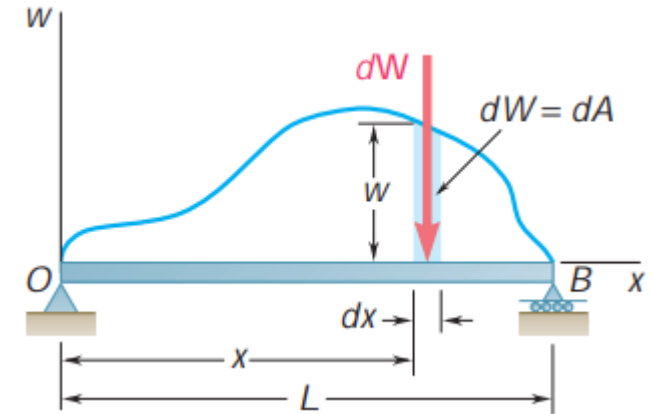
- Suppose if the beam is subjected to distributed load defined by $w(x)$. The total load W it imposed on the supports is given by:

$$W = \int_0^L w(x) dx$$

- When calculating the reaction forces at the supports, it is more convenient to represent this distributed load by an equivalent point force W that passes through its centroid, C .

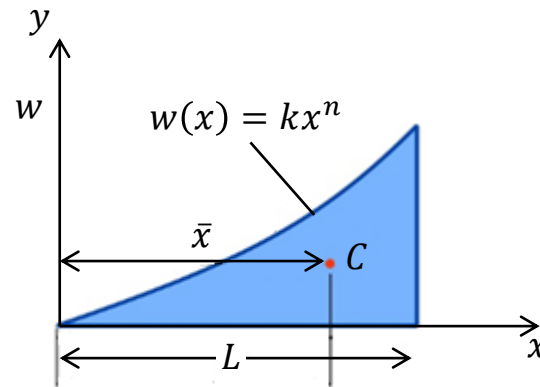
- Its x -coordinate $\bar{x}$ can be evaluated by

$$W\bar{x} = \int_0^L x \cdot w(x) dx \Rightarrow \bar{x} = \frac{\int_0^L x \cdot w(x) dx}{W}$$



Reaction Forces with Distributed Loads...

- Consider a general distributed load shown.



$$W = \int_0^L kx^n dx = k \frac{L^{n+1}}{n+1}$$

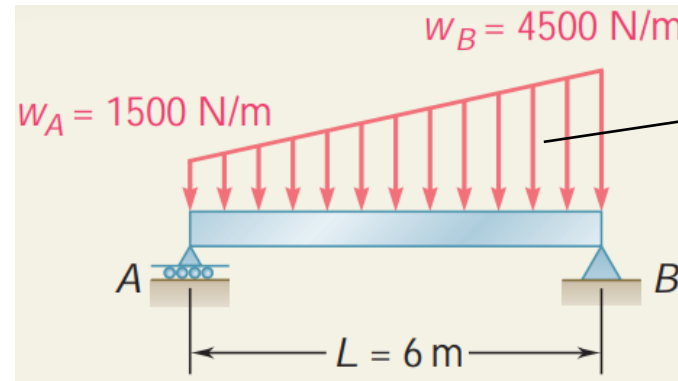
$$\int_0^L x(kx^n) dx = k \frac{L^{n+2}}{n+2}$$

$$\Rightarrow \bar{x} = \frac{k \frac{L^{n+2}}{n+2}}{k \frac{L^{n+1}}{n+1}} = \frac{n+1}{n+2} L$$

For linear distribution ($n = 1$), this gives the centroid of a triangle, i.e. $\bar{x} = \frac{2}{3} L$.

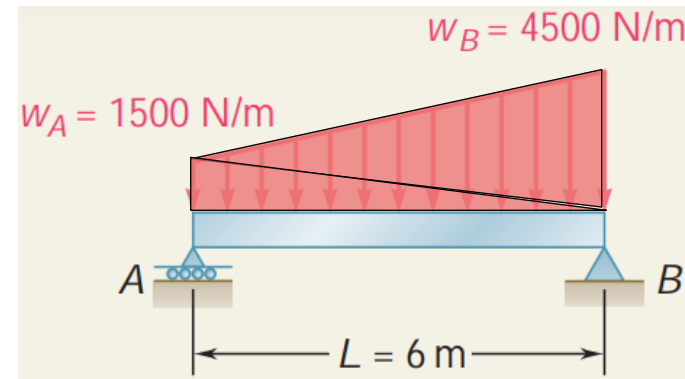
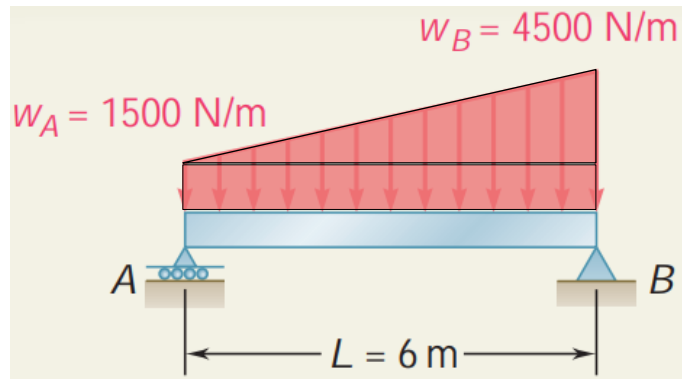
Example 5.4

A beam supports a distributed load as shown in the figure. Determine the reactions at the supports.

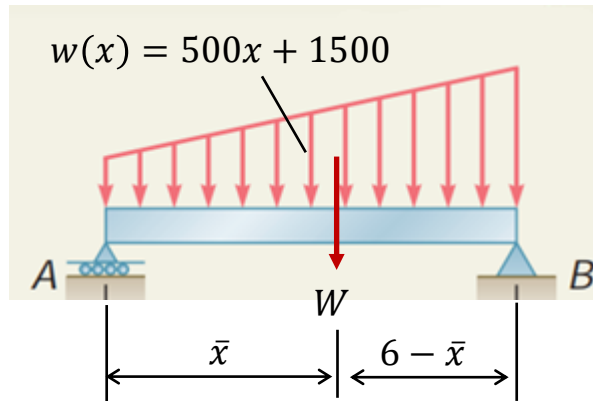


$$w(x) = \frac{(4500 - 1500)}{6}x + 1500$$

$$= 500x + 1500$$



Example 5.4...



Case 1:

Based on the distributed load function, we have

$$W = \int_0^L 500x + 1500 dx = 18000 \text{ N}$$

$$\int_0^L x(500x + 1500) dx = 63000 \text{ N}\cdot\text{m}$$

$$\bar{x} = \frac{63000}{1800} = 3.5 \text{ m}$$

Taking moment about A, gives

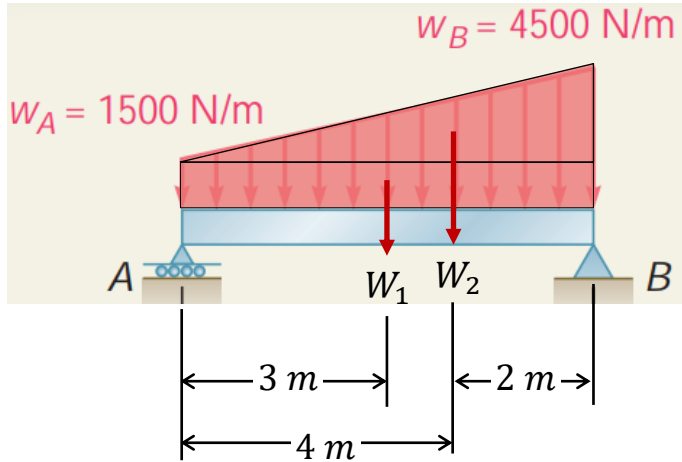
$$B(6) - W\bar{x} = 0 \Rightarrow B(6) - (18000)(3.5) = 0$$

$$\Rightarrow B = \mathbf{10500 \text{ N}}$$

Taking moment about B, we have

$$-A(6) + (18000)(6 - 3.5) = 0 \Rightarrow A = \mathbf{7500 \text{ N}}$$

Example 5.4...



Case 2:

Equivalent loads are:

$$W_1 = w_A L = 1500(6) = 9000 \text{ N}$$

And

$$W_2 = \frac{1}{2}(w_B - w_A)L$$

$$\Rightarrow W_2 = \frac{1}{2}(4500 - 1500)(6) = 9000 \text{ N}$$

Taking moment about A, we have

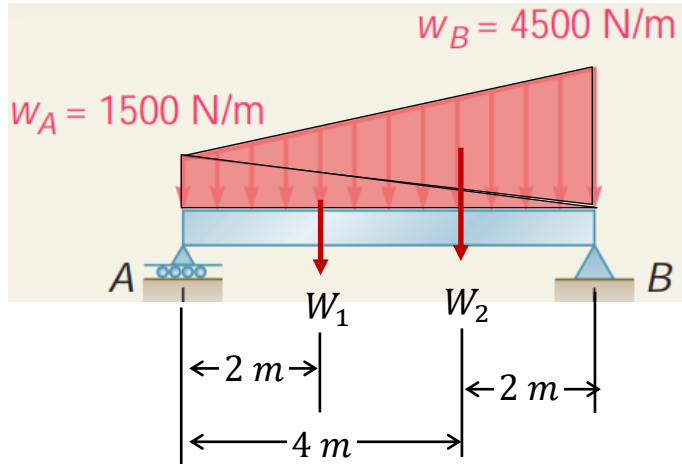
$$B(6) - W_1(3) - W_2(4) = 0 \Rightarrow B(6) - (9000)(3) - (9000)(4) = 0$$

$$\Rightarrow B = \mathbf{10500 \text{ N}}$$

Taking moment about B, we have

$$-A(6) + (9000)(3) + (9000)(2) = 0 \Rightarrow A = \mathbf{7500 \text{ N}}$$

Example 5.4...



Case 3:

Equivalent loads are:

$$W_1 = \frac{1}{2} (1500)(6) = 4500 \text{ N}$$

And

$$W_2 = \frac{1}{2} (4500)(6) = 13500 \text{ N}$$

With this, moment equations about A and B, gives

$$B(6) - (4500)(2) - (13500)(4) = 0$$

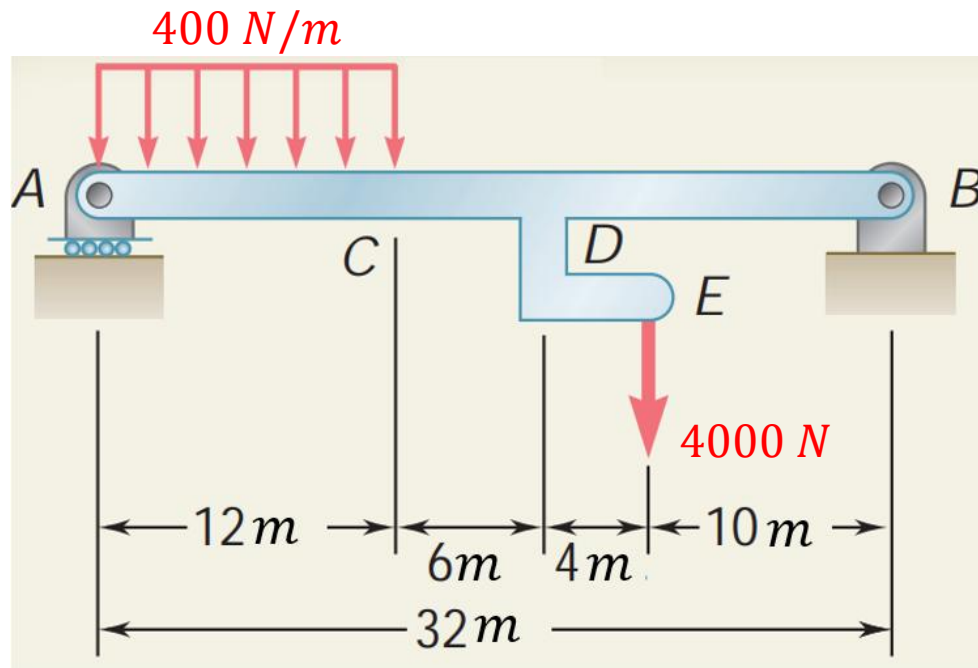
$$\Rightarrow B = \mathbf{10500 \text{ N}}$$

$$-A(6) + (4500)(4) + (13500)(2) = 0$$

$$\Rightarrow A = \mathbf{7500 \text{ N}}$$

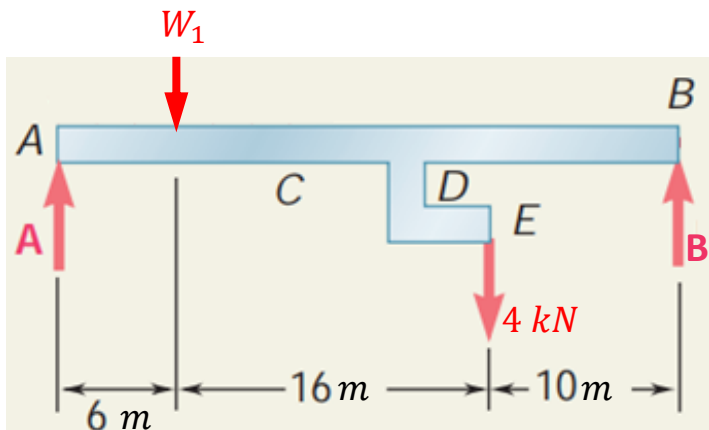
Example 5.5

Draw the shear and bending-moment diagrams for the beam AB . The distributed load of 400 N/m extends over 12 m of the beam, from A to C , and the 4000 N load is applied at E .



Example 5.5...

Consider the *FBD* of the beam.



The distributed load is replaced by its equivalent load,

$$W_1 = 400(12) = 4.8 \text{ kN}$$

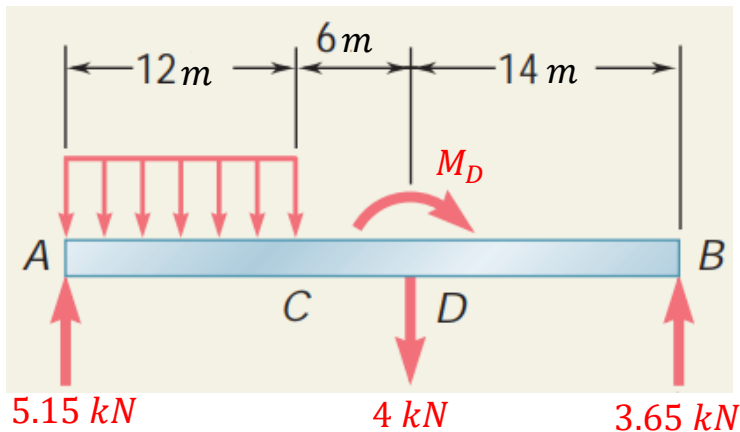
Taking moment about A, we have

$$-(4.8)(6) - 4(22) + B(32) = 0$$

$$\Rightarrow B = 3.65 \text{ N}$$

And vertical force equation gives

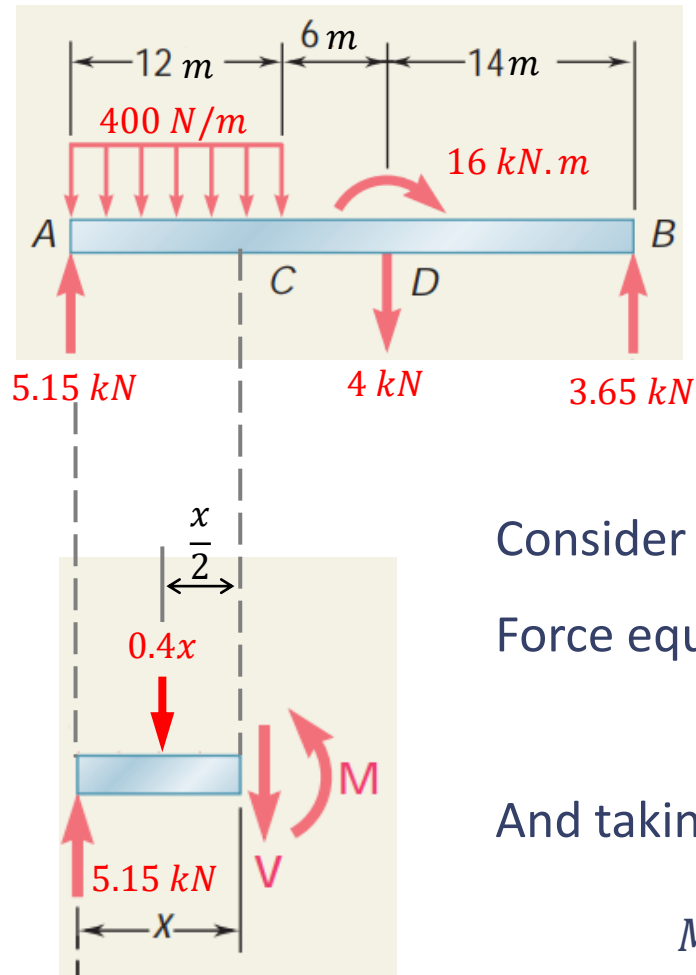
$$A + B - W_1 - 4 = 0 \Rightarrow A = 5.15 \text{ kN}$$



To analysis beam *AB*, the 4 *kN* force at *E* is replaced by its force-couple pair, with couple equals to

$$M_D = -4(4) = -16 \text{ kN.m}$$

Example 5.5...



Based on the loadings on the beam, we can identify 3 segments, i.e. *AC*, *CD* and *DB*.

Consider segment *AC*.

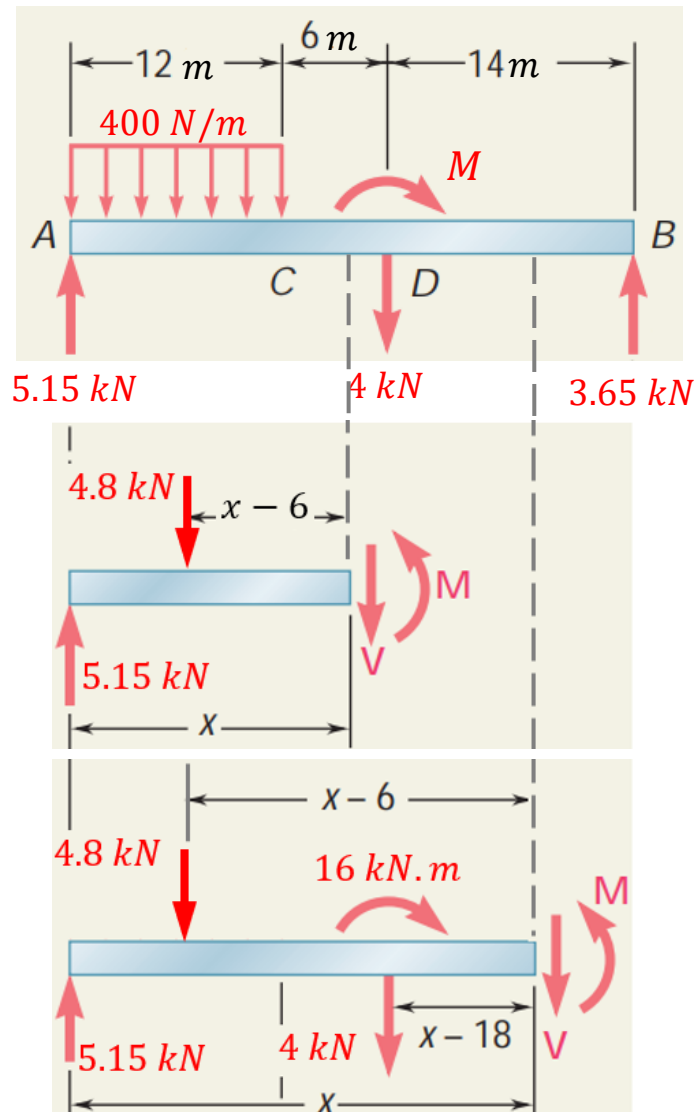
Force equation gives

$$5.15 - 0.4x - V = 0 \Rightarrow V = 5.15 - 0.4x$$

And taking moment about cut-section

$$M + 0.4x \left(\frac{x}{2} \right) - 5.15(x) = 0 \Rightarrow M = 5.15x - 0.2x^2$$

Example 5.5...



Next, consider segment *CD*.

Force equation gives

$$5.15 - 4.8 - V = 0 \Rightarrow V = 0.35 \text{ kN}$$

And taking moment about cut-section

$$M + 4.8(x - 6) - 5.15(x) = 0$$

$$\Rightarrow M = 0.35x + 28.8$$

Similarly, for segment *DB*.

Force equation gives

$$5.15 - 4.8 - 4 - V = 0 \Rightarrow V = -3.65 \text{ kN}$$

And taking moment about cut-section

$$M - 16 + 4(x - 18) + 4.8(x - 6) - 5.15(x) = 0$$

$$\Rightarrow M = 116.8 - 3.65x$$

Example 5.5...

Finally, with the shear force and bending-moment expressions, we can plot them as follows:

Shear force V :

$$AC \rightarrow V = 5.15 - 0.4x, \text{ for } 0 \leq x < 12$$

$$CD \rightarrow V = 0.35 \text{ kN}, \text{ for } 12 \leq x < 18$$

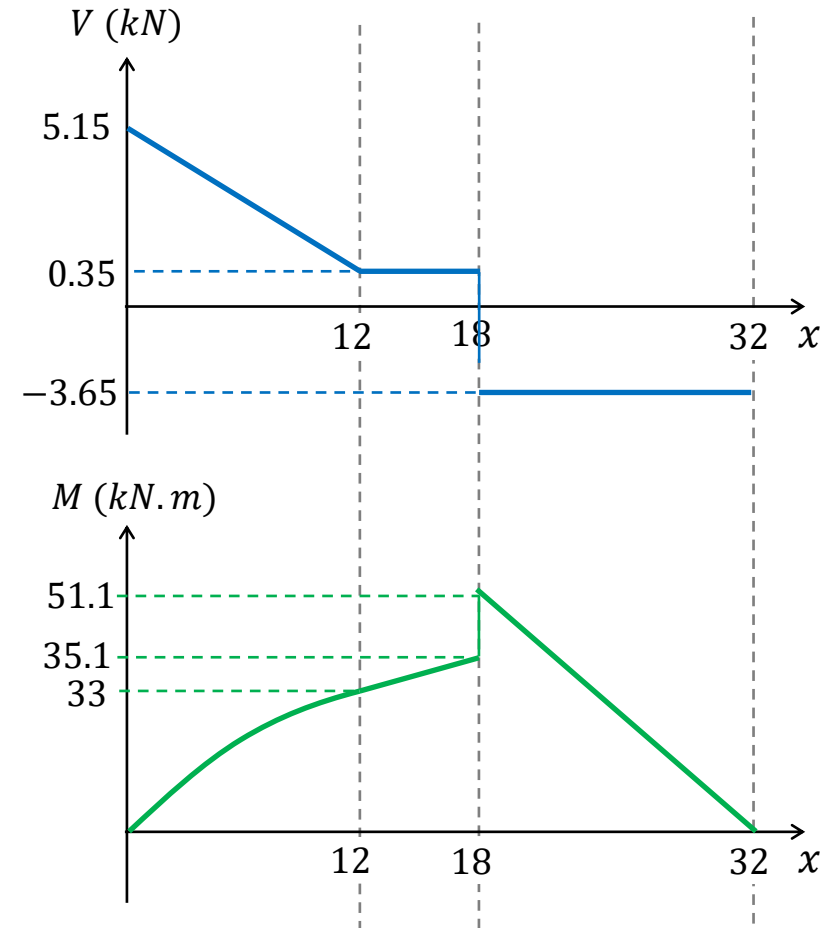
$$DB \rightarrow V = -3.65 \text{ kN}, \text{ for } x \geq 18$$

Bending-moment M :

$$AC \rightarrow M = 5.15x - 0.2x^2, \text{ for } 0 \leq x < 12$$

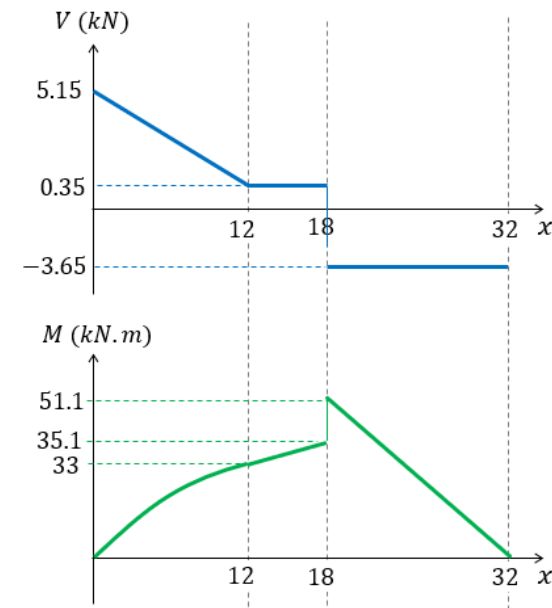
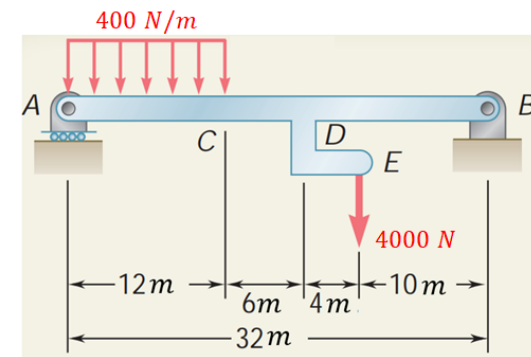
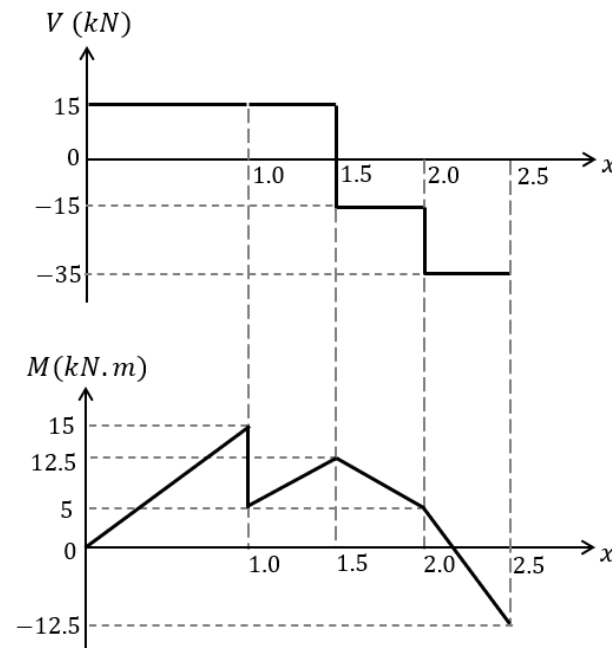
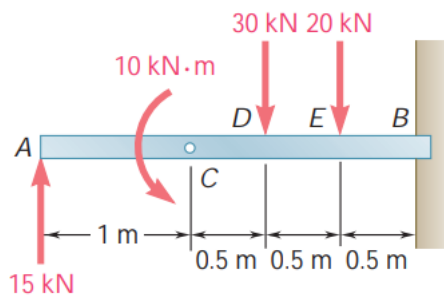
$$CD \rightarrow M = 0.35x + 28.8, \text{ for } 12 \leq x < 18$$

$$DB \rightarrow M = 116.8 - 3.65x, \text{ for } x \geq 18$$



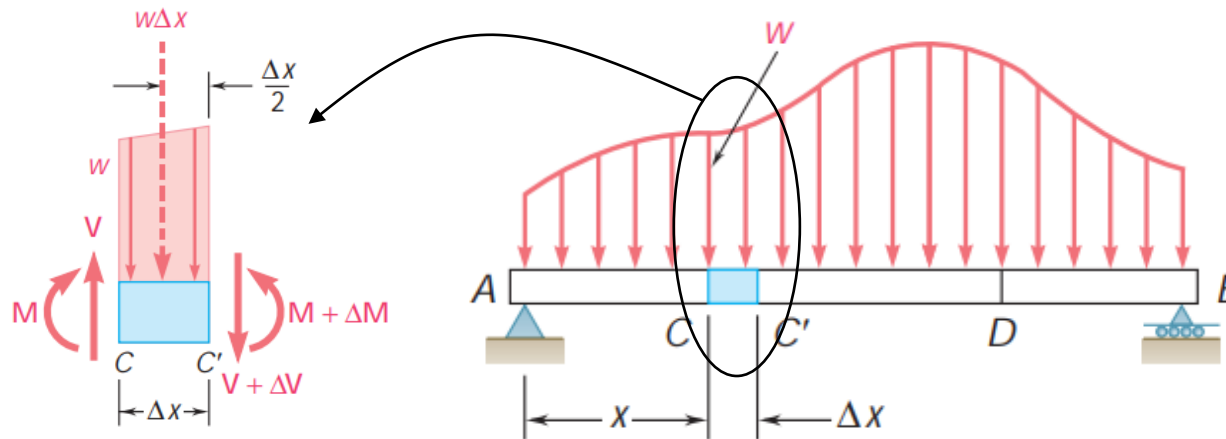
Relations among Load, Shear and Bending-Moment

- The above approach, although systematic, becomes cumbersome if the beam is subjected to complex loading profiles.
- Close examination of the shear and bending moment seems to suggest that there are some relationships among these variables.



Relations between Load and Shear

- Consider a beam that is loaded with a general distributed load w as shown.



- Suppose we isolate a small segment CC' of the beam. And from the force equilibrium in the vertical direction, we get

$$V - (V + \Delta V) - w\Delta x = 0 \Rightarrow \Delta V = -w\Delta x \Rightarrow \frac{\Delta V}{\Delta x} = -w$$

Now, in the limit as $\Delta \rightarrow 0$, the above expression becomes

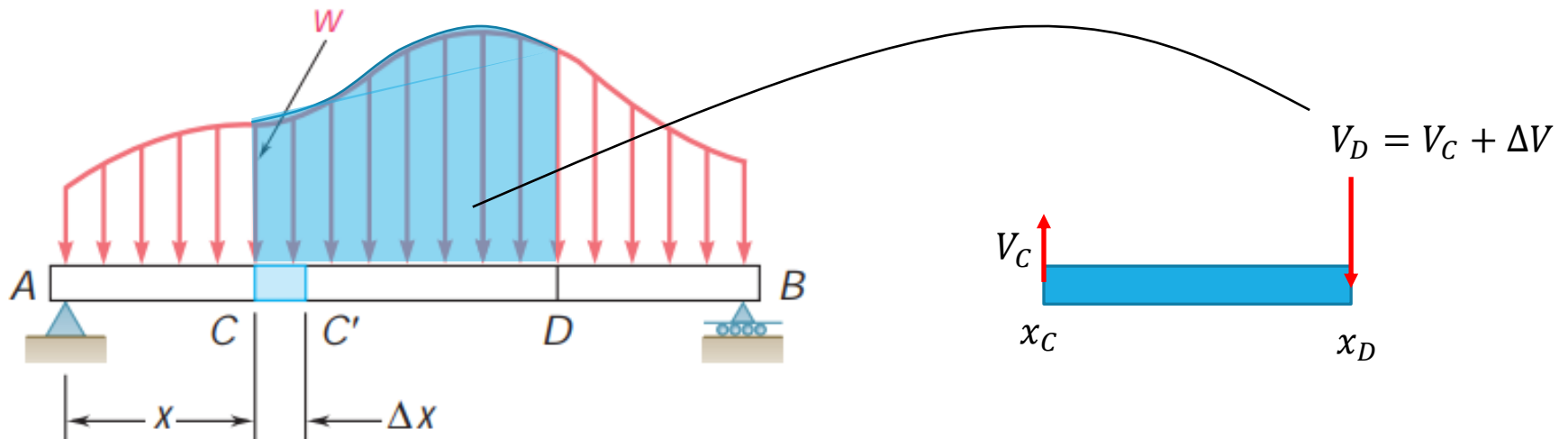
$$\frac{dV}{dx} = -w$$

Relations between Load and Shear...

- Hence, given a continuous shear force profile, the corresponding distributed loads w can be derived by differentiating of the shear force expression.
- On the other hand, this also means that by integrating the load function, we get

$$V_D - V_C = - \int_{x_C}^{x_D} w \, dx$$

$$V_D - V_C = -(\text{area under load curve between } C \text{ and } D)$$



Relations between Shear and Bending-Moment

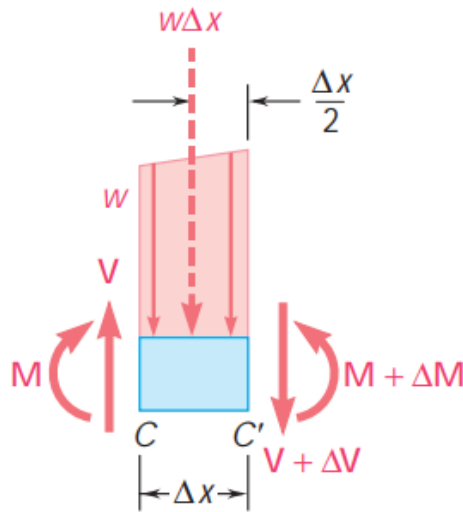
- Returning to the segment CC' and now taking moment about C' , we have

$$(M + \Delta M) - M - V\Delta x + w\Delta x \frac{\Delta x}{2} = 0$$

$$\Rightarrow \Delta M = V\Delta x - \frac{1}{2}w\Delta x^2$$

Again in the limit as $\Delta \rightarrow 0$, the above equation becomes

$$\frac{dM}{dx} = V$$

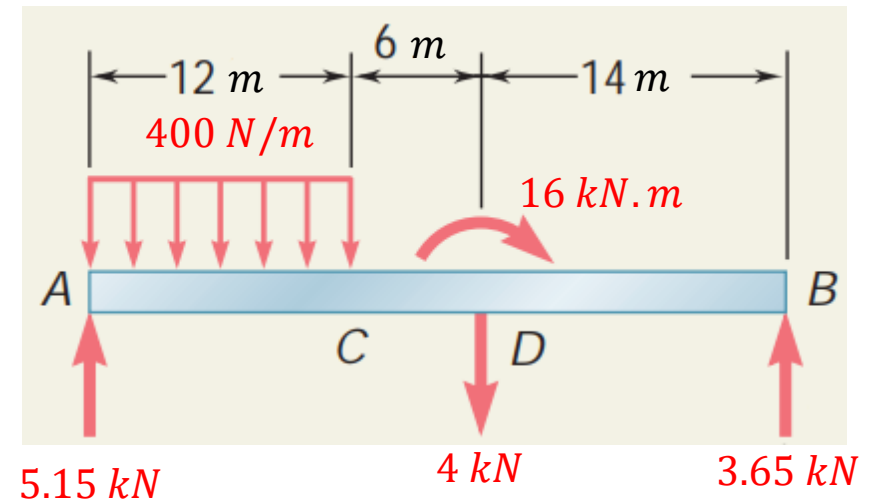
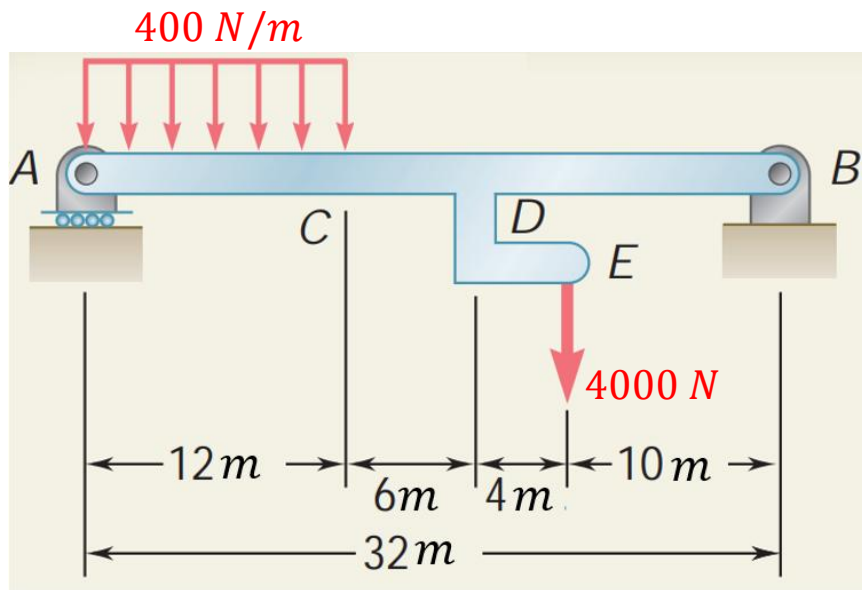


- Using the same concept as in the shear and load relations, it can also be deduced that

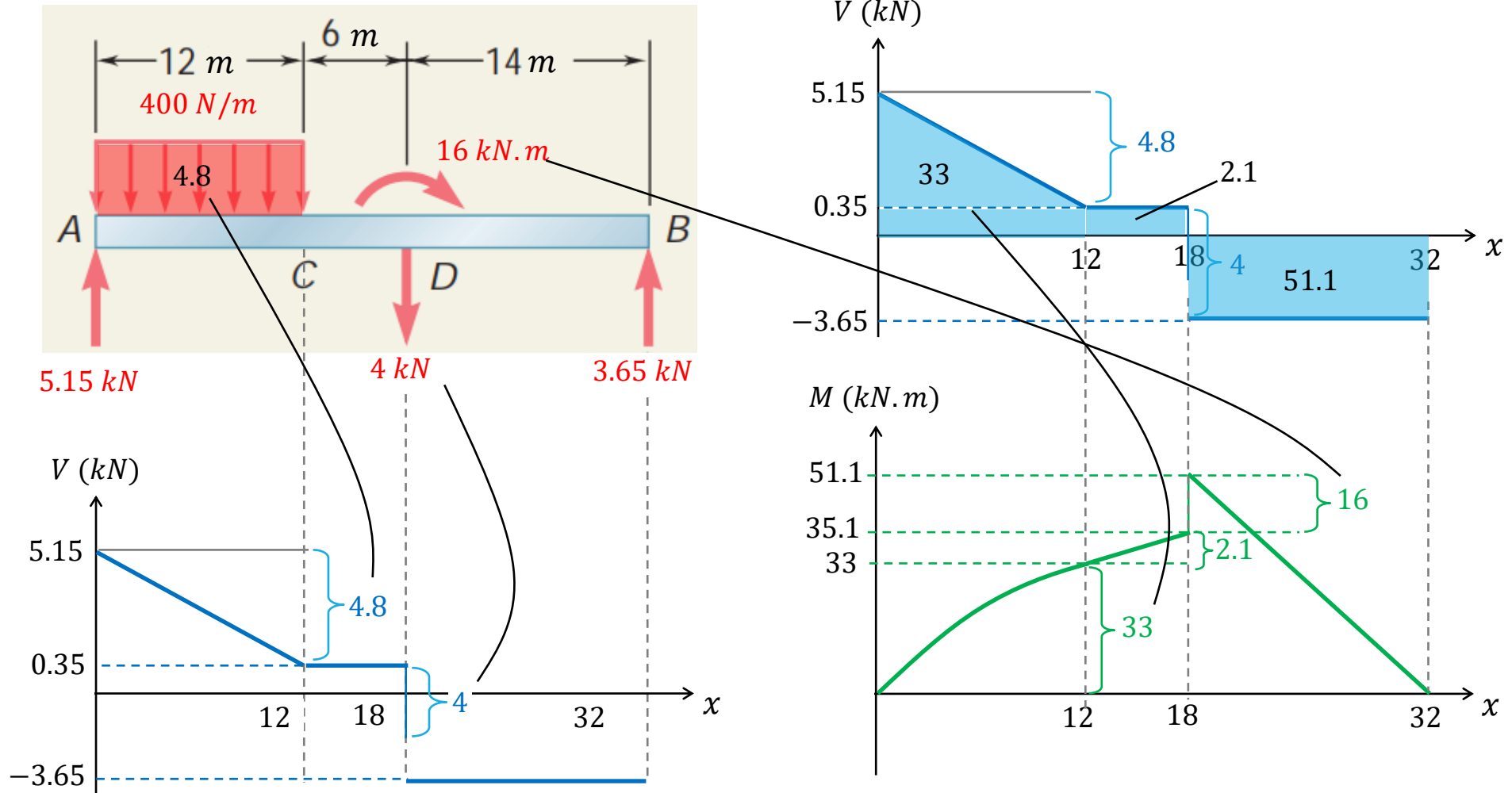
$$W_D - W_C = \int_{x_C}^{x_D} V dx = \text{area under shear curve between } C \text{ and } D$$

Example 5.6 (Redo Example 5.5)

Draw the shear and bending-moment diagrams for the beam AB . The distributed load of 400 N/m extends over 12 m of the beam, from A to C , and the 4000 N load is applied at E .

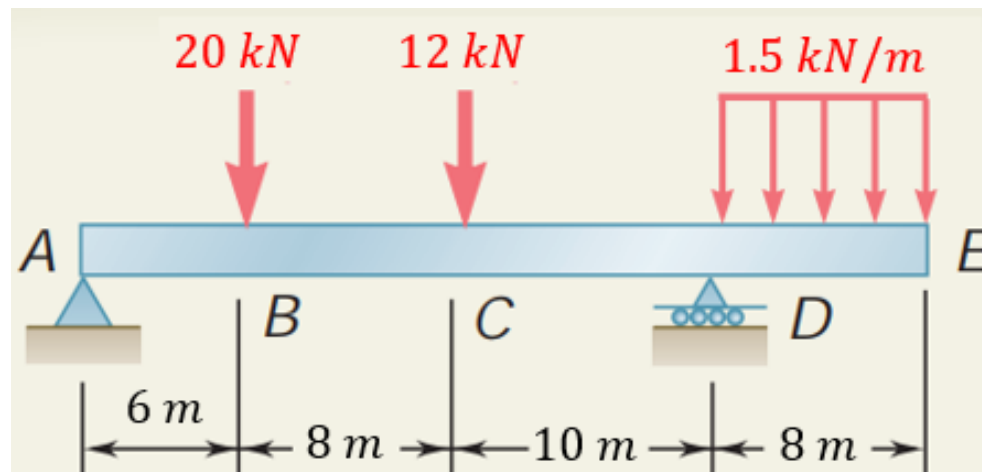


Example 5.6...



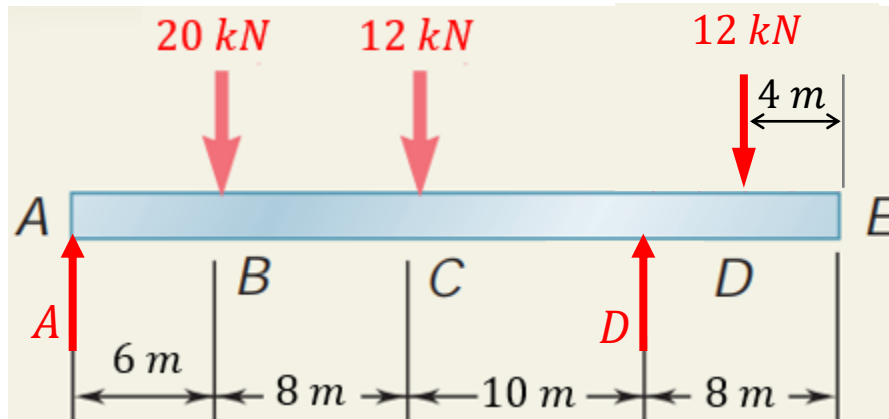
Example 5.7

Draw the shear and bending-moment diagrams for the beam and loading shown and hence determine the largest shear force and bending-moment, and their corresponding locations.



Example 5.7...

Consider the *FBD* of the entire beam.



First, the distributed load between *DE* is equivalent to a point force of 12 *kN* applied at 4 *m* from *E*.

Taking moment about A, gives

$$-20(6) - 12(14) + D(24) - 12(28) = 0 \Rightarrow D = 26 \text{ kN}$$

And force equilibrium vertically, gives

$$A - 20 - 12 + 26 - 12 = 0 \Rightarrow A = 18 \text{ kN}$$

Example 5.7...

